

Funding Proposal

FP196: Supporting Innovative Mechanisms for Industrial Energy Efficiency Financing in Indonesia with Lessons for Replication in other ASEAN Member States

Indonesia | Korea Development Bank (KDB) | Decision B.34/10

14 November 2022



Funding Proposal

Project/Programme title:	Supporting Innovative Mechanisms for Industrial Energy Efficiency Financing in Indonesia with Lessons for Replication in other ASEAN Member States
Country(ies):	Indonesia
Accredited Entity:	The Korea Development Bank
Date of first submission:	<u>[2021/08/26]</u>
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GREEN
CLIMATE
FUND

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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

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Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

List of Acronyms

ACE	ASEAN Centre for Energy
AE	Accredited Entity
AMA	Accreditation Master Agreement
AMS	ASEAN Member States
APAEC	ASEAN Plan for Action for Energy Cooperation
ASEAN	Association of Southeast Asian Nations
BAPPENAS	Ministry of National Development Planning of the Republic of Indonesia
BASE	Basel Agency for Sustainable Energy
BAU	Business-as-usual
BKF	Fiscal Policy Agency (Badan Kebijakan Fiskal)
CAPEX	Capital Expenditures
CDM	Clean Development Mechanism
COVID-19	Coronavirus Disease
CSO	Civil and Social Organisation
DFI	Development Financial Institution
E&S	Environmental and Social
EE	Energy Efficiency
EE&C	Energy Efficiency and Conservation
EEF	Energy Efficiency Financing
ESCO	Energy Service Company
ESG	Environmental, Social, and Governance
ESI	Energy Savings Insurance
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
ESS	Environmental and Social Safeguards
FAA	Funded Activity Agreement
FI	Financial Institution (or Financial Intermediary)
GAP	Gender Action Plan
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GRC	Grievance Redressal Committee
IE	Implementing Entity
IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
JCM	Joint Credit Mechanism
KDB	The Korea Development Bank
KEA	Korea Energy Agency
KKP	Ministry of Maritimes Affairs and Fishery of Indonesia
LFIs	Local Financial Intermediaries
MDB	Multilateral Development Bank
MEPS	Minimum Energy Performance Standard
NDA	National Designated Authority
NDC	Nationally Determined Contribution
O&M	Operation and Management



OECD	Organisation for Economic Co-operation and Development
OJK	Financial Services Authority (Otoritas Jasa Keuangan)
OPEX	Operating Expenses
PIU	Programme Implementation Unit
PLN	State Electricity Company (Perusahaan Listrik Negara)
PoA	Programme of Activities
PT IIF	PT Indonesia Infrastructure Finance
PT SMI	PT Sarana Multi Infrastruktur
RCP	Representative Concentration Pathway
RPJMN	National Medium Term Development Plan (Narasi Rencana Pembangunan Jangka Menengah Nasional)
RUEN	National Energy General Plan (Rencana Umum Energi Nasional)
RUPTL	National Electricity Supply Business Plan (Rencana Usaha Penyediaan Tenaga Listrik)
SDGs	Sustainable Development Goals
SIAC	Singapore International Arbitration Centre
SME	Small and Medium-sized Enterprise
TA	Technical Assistance
tCO ₂ eq	Tonne of Carbon dioxide (CO ₂) equivalent
TCX	Currency Exchange Fund
TOE	Tonnes of Oil Equivalent
UNFCCC	United Nations Framework Convention on Climate Change
VFD	Variable Frequency Drive
VSD	Variable Speed Drive
WHR	Waste Heat Recovery

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Private
A.3. Request for Proposals (RFP)	<u>Not applicable</u>		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.		
		GCF contribution	Co-financers' contribution¹
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Energy generation and access	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %
	☒ Buildings, cities, industries and appliances	100 %	100 %
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Most vulnerable people and communities	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Health and well-being, and food and water security	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Infrastructure and built environment	<u>Enter number</u> %	<u>Enter number</u> %
☐ Ecosystems and ecosystem services	<u>Enter number</u> %	<u>Enter number</u> %	
A.5. Expected mitigation outcome <i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>	3,139KtCO ₂ eq over Programme Total Life Span	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	-
			-
			-
A.7. Total financing (GCF + co-finance²)	247.7 million USD	A.9. Project size	Medium (Upto USD 250 million)
A.8. Total GCF funding requested	<u>105 million</u> USD		

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e., GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant 5 million US\$ million USD ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☒ Guarantee 100 million USD		
A.11. Implementation period	10 years	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	1/20/2023	A.14. ESS category	I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☐ No ☒		
A.20. Executing Entity information	For Component 1 Local Financial Intermediaries (Local FIs or LFIs) will be engaged as Executing Entities, i.e., financial intermediaries, as per local stakeholders' interest and appetite. - PT Sarana Multi Infrastruktur (SMI) - PT Indonesia Infrastructure Finance (IIF) - PT KEB Hana Indonesia (Hana Indonesia) - PT Bank IBK Indonesia (IBK Indonesia) - KB Bukopin (KB Indonesia) - PT Bank Shinhan Indonesia (Shinhan Indonesia). - PT Bank Woori Saudara Indonesia (Woori Indonesia) For Components 2 and 3 - ASEAN Centre for Energy (ACE) A Jakarta-based intergovernmental organisation within the ASEAN structure which represents the 10 ASEAN Member States (AMS) interests in the energy sector		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

Indonesia is in need of urgent actions to address gradually deteriorating climate change impacts and their consequences. The climate-vulnerable archipelagic state submitted its nationally determined contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC) with a strong willingness to transition towards a low-carbon and climate-resilient pathway. The NDC of Indonesia recognises energy efficiency as one of its key mitigation actions. While Indonesia attributes 65.43% of the total greenhouse gas (GHG) emissions to the energy sector, the market penetration of energy efficient technologies and the ensuing energy savings have been long hampered by a range of barriers, such as access to finance, regulatory challenges, and a lack of capacity and awareness amongst relevant stakeholders. In addition, since the Coronavirus Disease (COVID-19) outbreak and spread, the state budget and relevant stimulus measures in the neediest areas have brought the agenda of energy efficiency and conservation back in terms of priority. Surrounded by further setbacks from the pandemic and related budget deficits, an international-level intervention in the urgent, but deprioritised area is more important than ever. The Programme entitled “Supporting Innovative Mechanisms for Industrial Energy Efficiency Financing in Indonesia with Lessons for Replication in the other ASEAN Member States,” hereafter “Programme,” will support Indonesia and its neighbouring ASEAN members to re-focus on climate actions amid the post-pandemic recovery era, strengthening the foundation for the NDC achievement and the national alignment with the Paris Agreement eventually.

The Programme, therefore, shall support Indonesia and other ASEAN countries to be ready to drive a low-carbon development pathway with enhanced energy efficiency and conservation performance, addressing three chronic barriers – i.e., financial, regulatory, and technical barriers, which are all interlinked with a vicious circle of implementation failures of similar precedents. This initiative pays special attention to the three challenges, regarding them as the three entry points of intervention and designing three components accordingly as below. The Programme proposes a solution package of innovative energy efficiency financing schemes equipped with de-risking mechanisms, a supportive regulatory framework, and technical assistance.

[Table 1] Programme Composition Summary

Triggered Barrier	Component	Executing Entity	GCF Funding
<i>Financial barrier (e.g., access to energy efficiency finance)</i>	[Component 1] Energy Efficiency Finance	Local financial intermediaries (FIs)	USD100 mil (Guarantees)
<i>Regulatory barrier (e.g., lack of policy incentives)</i>	[Component 2] De-risking mechanisms	ASEAN Centre for Energy (ACE)	USD1.58 mil (TA grant)
<i>Demand-side barrier (e.g., weak capacity of local enterprises)</i>	[Component 3] Technical Assistance (TA) for the Market Readiness	ASEAN Centre for Energy	USD3.42 mil (TA grant)

The Programme anticipates bringing about a transformative change in the industrial energy efficiency market of Indonesia and its neighbouring ASEAN countries, in terms of the GHG emissions reduction (3,139,000 tonnes of CO₂eq), as well as proliferation of diverse energy efficient technologies, socio-economic and environmental co-benefits, investment at scale, awareness enhancement and behavioural change amongst local players, and so on. Above all things, the Programme will function as a catalyst to drive a genuine paradigm shift through enabling environment formation, in addition to one-off financing provision. This grand-scale intervention will be made with key local and global stakeholders with different expertise under KDB supervision. With the successful implementation of financing under component 1 and TA activities under component 3, i) confidence over energy efficient technologies will be accumulated amongst industries and financiers on the ground, ii) financiers will have a better understanding on energy efficiency financing and a credit analysis and mitigation mechanisms within their internal system, and iii) energy efficiency ecosystem including locally fit de-risking scheme and regulatory framework development will be created in Indonesia industry sector, which will make financial institutions and local stakeholders keep joining energy efficiency financing and market even after the Programme is over.

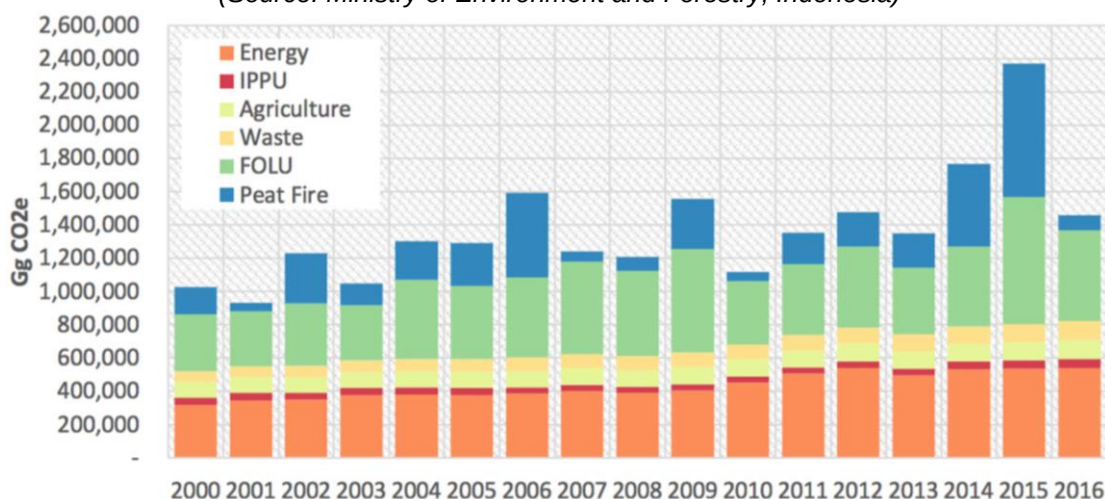
B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Mitigation Needs in the Energy Sector of Indonesia, Surrounded by Harsh Market Circumstances. Indonesia is one of the largest GHG emitters globally. The Carbon Brief Profile (2019) recognises the large emerging economy as the world's fourth largest GHG emitter in 2015, attributing heavy emissions to the burning of fossil fuels for energy generation, deforestation, and peat fires. In particular, the energy sector largely contributes to the total GHG emissions of the large emerging economy. Relevantly, Climate Transparency (2020) argues that Indonesia has seen its emissions increase by almost 140% between 1990-2017 with the highest increase in the energy sector. A business as usual (BAU) scenario-based emissions project presents a dire snapshot of the energy sector being the biggest culprit contributing significantly to the enormous GHG emissions in 2030 (BAPPENAS, 2015). Given this backdrop, Indonesia needs urgent and effective actions to mitigate GHG emissions in the energy sector in pursuit of a true transformation towards a low-carbon society.

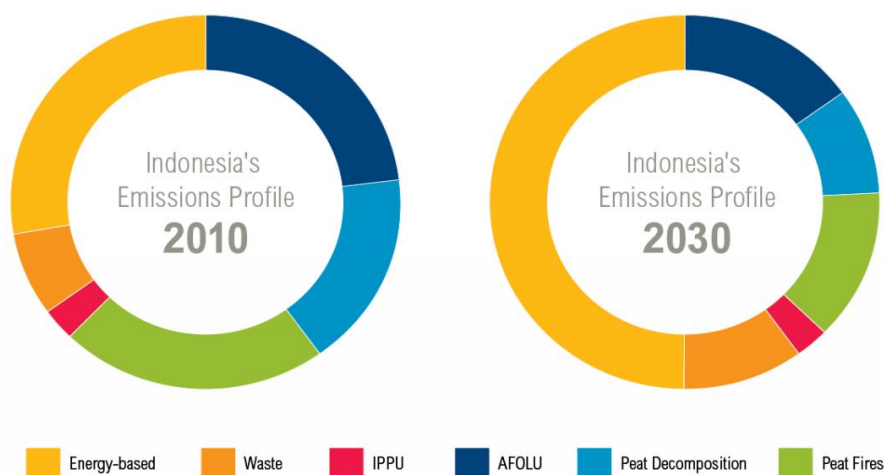
【Figure 1】 GHG Emissions Profile of Indonesia, 2000-2016

(Source: Ministry of Environment and Forestry, Indonesia)



【Figure 2】 GHG Emissions Profile of Indonesia, 2010 vs. 2030

(Source: BAPPENAS, 2015, extracted from the World Resources Institute)

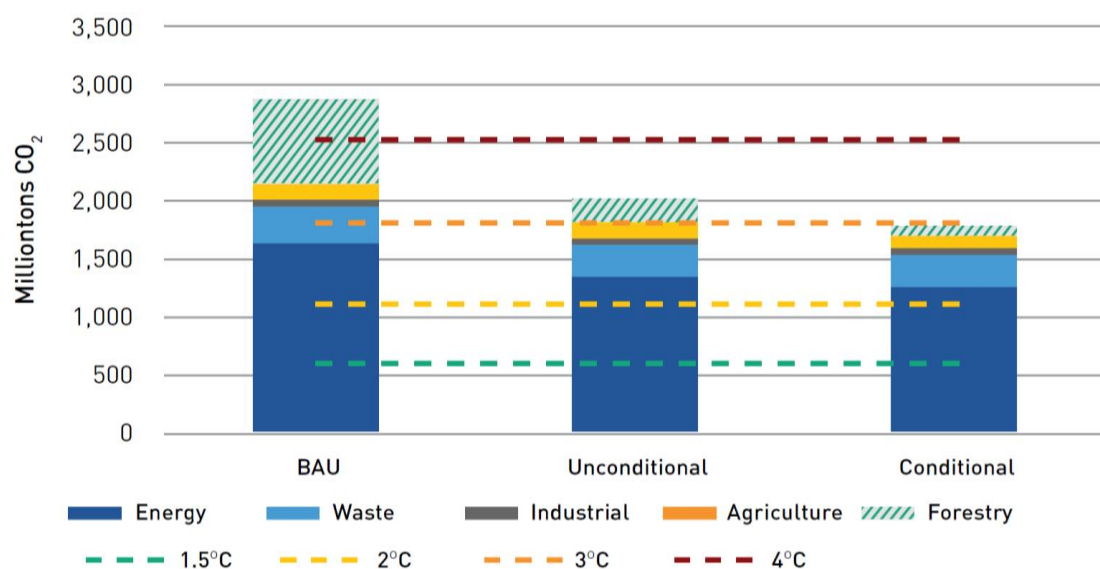


Indonesia's energy sector has faced challenging circumstances against a low-carbon transformation, considering the apparent fossil fuel dependency; as of 2020, more than 60% of its electricity generation relied on the capacity of coal-fired power plants, according to the latest Power Development Plan (RUPTL)

published by the state-owned utility company, PLN. Such a heavy dependency has been continuously exacerbated with the state subsidies for the fossil fuel-based petroleum products and electricity. As of 2019, the amount of the fossil fuel subsidies totalled USD8.6 billion. Distorted policy incentives have resulted in significantly low energy prices, which is one of the critical barriers to energy transition moves including the promotion of energy efficiency and conservation in Indonesia.

Intervention is Urgent and Necessary with Indonesia, Lagging behind the Paris Agreement. According to the NDC of Indonesia, updated in July 2021, the country has set up an unconditional target of 29% reduction of the GHG emissions against the BAU scenario by 2030, while committing itself to a conditional contribution up to 41% reduction by 2030 with international support. In actuality, these targets are criticised by international experts because they are not ambitious enough to be in line with the Paris Agreement. While the Ministry of Environment and Forestry of Indonesia also reset the net zero target year from 2070 to 2060 through the updated NDC, it emphasises the necessity of international assistance. Given that the national net zero target of 41% conditional mitigation commitment by 2060 can be achieved only with internationally supportive actions for finance provision, technology development and transfer, and capacity building, an international level intervention is critical. At this juncture, the international partnership-based Programme is an essential intervention for Indonesia. It shall tackle the energy sector with substantial mitigation potential (see below) for the country to be on track moving towards a 1.5°C scenario.

【Figure 3】 Indonesia's Updated NDC Scenario in 2030

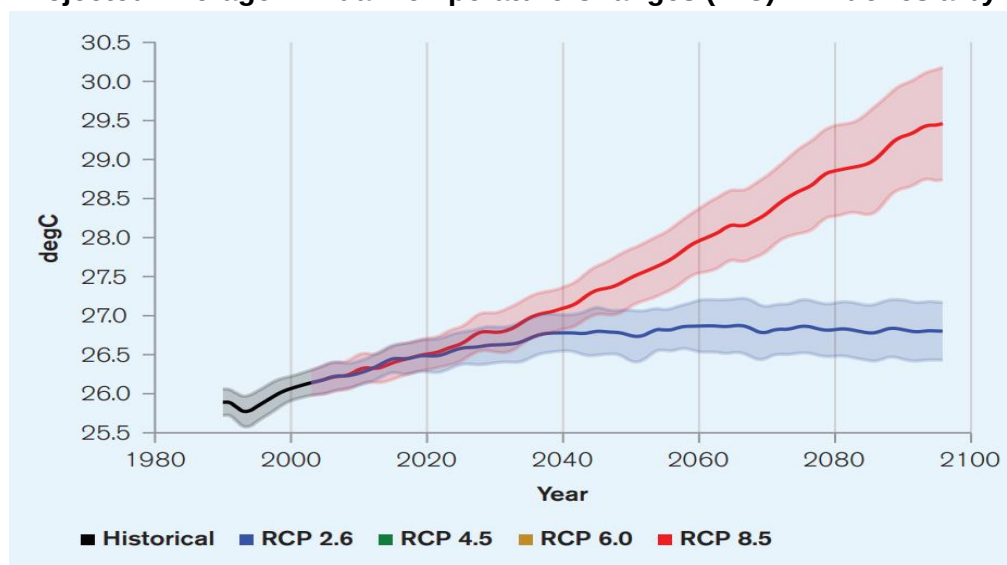


Internationally projected climate scenarios present a dire picture of the climate future in case of a BAU pathway; Indonesia is not an exception to this at all. In particular, Indonesia's scenarios of a Representative Concentration Pathway (RCP), a GHG concentration trajectory adopted by the Intergovernmental Panel on Climate Change (IPCC), alarm the country when it trails around with a BAU case of RCP8.5. The global community ought to support Indonesia to strive for and act in accordance with a choice of RCP2.6, a desirable pathway that limits global warming to below 1.5°C, the aspirational goal of the Paris Agreement. See below RCP scenarios for Indonesia, by different emissions pathways from the strong mitigation scenario to the BAU scenario.

【Table 2】 Projected Temperature Changes (° C) in Indonesia by RCP Scenario

Scenario	Average Daily Maximum Temperature		Average Daily Temperature		Average Daily Minimum Temperature	
	2040~2059	2080~2099	2040~2059	2080~2099	2040~2059	2080~2099
RCP2.6	0.9 (0.2, 1.5)	0.9 (0.2, 1.8)	0.9 (0.4, 1.5)	0.9 (0.3, 1.6)	0.9 (0.4, 1.5)	0.9 (0.3, 1.5)
RCP4.5	1.2 (0.5, 2.0)	1.6 (0.8, 2.6)	1.2 (0.7, 1.8)	1.6 (1.0, 2.5)	1.2 (0.7, 1.8)	1.7 (1.1, 2.5)
RCP6.0	1.0 (0.3, 1.9)	2.0 (1.2, 3.0)	1.1 (0.6, 1.7)	2.0 (1.4, 2.9)	1.1 (0.6, 1.7)	2.0 (1.4, 3.0)
RCP8.5	1.6 (0.8, 2.5)	3.4 (2.4, 4.9)	1.6 (1.1, 2.3)	3.4 (2.6, 4.6)	1.6 (1.1, 2.3)	3.4 (2.6, 4.6)

【Figure 4】 Projected Average Annual Temperature Changes (° C) in Indonesia by RCP Scenario



Energy Efficiency is an Urgent Area for Reducing GHG Emissions from Indonesia's Energy Sector.

The International Energy Agency (IEA) estimates that half of the worldwide climate mitigation actions must come from energy efficiency.³ According to the IEA report on Southeast Asia's Energy Transition as of 2019, there is a significant untapped potential to improve energy efficiency measures in the industry sector across the Southeast Asian region and more than USD 600 billion is needed to improve end-use energy efficiency in the Sustainable Development Scenario that outlines the needed changes of the global energy system to deliver on these three energy-related Sustainable Development Goals (SDGs) – (1) universal access to energy (SDG 7), (2) reduction of the severe health impacts of air pollution (SDG 3), and (3) climate action (SDG 13) – by 2040.⁴

The 5th ASEAN Energy Outlook addresses that, between 2015 and 2040,⁵ the total primary energy supply of Indonesia will be more than double in the absence of enhanced energy efficiency. The IEA strongly supports the urgent need of improved industrial energy efficiency, noting that the role of energy efficiency is crucial in order to achieve energy intensity reduction targets of 17% lower than BAU projections proclaimed under the National Energy General Plan of Indonesia (RUEN).⁶ Of the energy-related GHG emissions, particularly, the industrial sector is the largest driver within the energy areas, accounting for 37% of the total, followed by the transport sector (27%) and power sector (27%). Climate Transparency Report (2020) states,

³ IEA. 2017

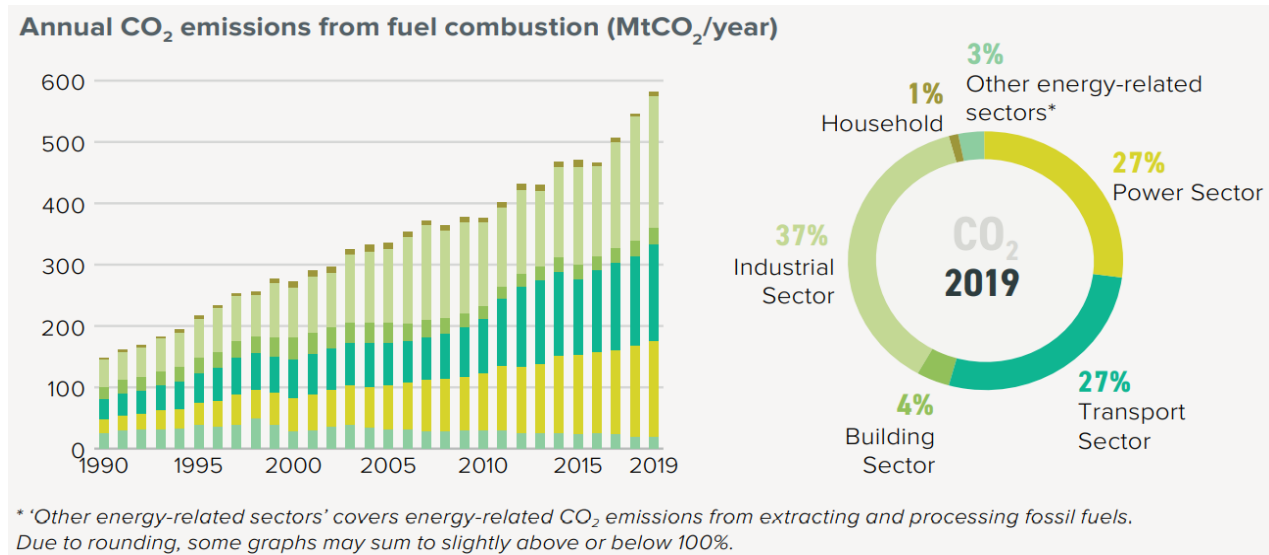
⁴ IEA. 2019. Southeast Asia Energy Outlook 2019

⁵ ASEAN Centre for Energy. 2017. The 5th ASEAN Energy Outlook (AEO5)

⁶ IEA. 2017

“Indonesia has implemented industrial energy management policies; however, more stringent industrial energy efficiency standards are needed to achieve significant emission reduction (p.13).” The sectoral breakdown of the GHG emissions justifies the Programme’s intervention in the industrial energy efficiency area by urgency and priority.

【Figure 5】 Energy-related GHG Emissions of Indonesia



Industrial Energy Efficiency in Indonesia – Potentials and Challenges. In emerging economies, in general, the energy use from industries is increasing exponentially, directly linked to the GDP. Between 2000 and 2015, Indonesia’s GDP has doubled and its demand for electricity increased by 150%.⁷ According to the ACE, energy consumption in the industry reached 11.94 million tonnes of oil equivalent (TOE) in 2017, which is around 30% of the total energy consumption in Indonesia. These figures highlight the importance of taking industrial energy efficiency measures in this climate vulnerable country. In addition, the GHG emissions from industrial activities that include fossil fuel combustions, industrial processes and products used, and waste treatment activities rank second after land use in Indonesia. Amongst manufacturing industries, the eight highest energy intensive industries – i.e., cement, fertilizer, ceramic and glass, chemical, pulp and paper, textile, iron and steel, and food and beverage – contributed approximately 55.9 percent of the national GHG emissions from energy use in industry (approx. 147.43 Mt CO₂e) as direct and indirect emissions in 2016.⁸ Severe inefficiencies in the industrial process and system have been leading to high energy waste on a national scale. The utilisation of energy-efficient technologies in the industry sector therefore could provide a substantial benefit to the GHG emissions reduction as well as to an energy intensity reduction target of Indonesia. In particular, potential energy savings would be significant in energy-intensive manufacturing industries where energy intensity could be improved by 44% by 2040.⁹

The Indonesian energy efficiency market, albeit a growth opportunity as suggested earlier, remains yet at the early development phase. The Market Assessment conducted by the Programme has recognised that only one initiative, the Joint Credit Mechanism (JCM), is operational out of numerous initiatives of competitive players – both domestic (public sector players) and global parties (multilateral and bilateral agencies) – so far to establish energy efficiency finance mechanisms in Indonesia. In addition, public incentives to spur the growth of the energy efficiency market are still limited. In 2017, Otoritas Jasa Keuangan (OJK), the Financial

⁷ International Energy Agency (IEA). 2017. Energy Efficiency 2017

⁸ Ministry of Industry Republic of Indonesia and UNDP. 2018. Greenhouse Gas Emission Profile from Indonesia’s Industry Sector

⁹ IEA. 2019

Services Authority, stipulated the OJK Regulation No.51/2017 regarding Sustainable Finance that obligated banks to apply sustainable finance principles and to submit yearly reports. Thus, financial institutions are legally required to provide financial products, such as loans for sustainable and green projects incorporating energy efficiency finance. However, since the regulation did not set any specific targets, financing towards energy efficiency efforts has not been accelerated. The nationwide deployment of technologies with a high energy efficiency potential in the Indonesian industry area has been hindered due to several challenges described in Table 3 below.

【Table 3】 Challenges in the Industrial Energy Efficiency Market in Indonesia

Financial Barriers	- Local financial institutions are unfamiliar with the energy efficiency financing mechanism with persistent implementation failure of precedents. - Banks are highly risk-averse in energy efficiency financing, thereby imposing high interest rates and asking a borrower for providing stringent credit and/or collateral and high equity injection which local SMEs are remotely capable of clinging to. - No credit mitigation technique including the de-risking mechanism (such as guarantee or insurance) for energy efficiency in the local market. - Financial institutions, in particular large-sized banking institutions, have little interest in financing energy efficiency projects since many are relatively small-scale projects led by SMEs with low credit. - High interest rates or collateral requirements for energy efficiency projects due to risk analysis difficulties.
Regulatory Barriers	- The subsidised energy tariff is a disincentive for industries to invest in energy savings; the price of electricity is U\$ 0.078/kWh for businesses (medium voltage),¹⁰ which is lower than that of other ASEAN Member states.¹¹ As part of the COVID-19 recovery measure, an incentive of 100% (later reduced to 50%) discount on electricity was provided, especially for low-income households and small businesses. - No minimum energy performance standard (MEPS) for industrial equipment and appliance is available to serve as guidance. - No regulation to encourage less energy intensive sectors (motor, boiler, etc.) due to lack of awareness amongst policy makers, despite the large GHG emission from those sectors. - Existing fiscal or non-fiscal incentives from the government to promote the energy efficiency area have not been disseminated to industries or financiers, nor been sufficient enough to boost the market. For instance, Article 20 of Government Regulation No.70/2009 (Energy Conservation) states that incentives may vary in the form of provision from taxation facility for energy saver equipment to low interest-rate funds for the need of investment in energy conservation. It, however, does not work in the market.
Demand-side Barriers	- Low demand for high-energy efficiency facilities due to low energy tariffs. - Market players lack awareness of assessing energy efficiency technologies and capacity and resources in carrying out its cost-benefit analysis, which partially results in a low prioritisation of investing in energy efficient projects. - Industries are yet to recognise the regulatory requirements with respect to energy efficiency reporting and implementation. - There are not many well-trained in-house energy managers nor extensive pools of experienced experts in energy efficiency, mainly due to little

¹⁰ <https://www.esdm.go.id/en/media-center/news-archives/electricity-tariffs-for-april-june-2021-remain-unchanged>

¹¹ Electricity prices for ASEAN member states: U\$ 0.113/kWh (Philippines), U\$ 0.11/kWh (Thailand), U\$ 0.092 (Malaysia) (1Q of 2021, Bloomberg NEF)

emphasis on the importance and distribution of resources and incentives for employing and training them.

As a result of multi-layered deteriorating dynamics amongst the aforementioned challenges and barriers, energy efficiency in Indonesian industries has not been facilitated and the relevant business models are not persuasive in the market. Given this, market players and stakeholders such as industries, technology providers, energy service companies (ESCOs), and financiers are not fully ready to engage in the deployment of energy efficiency technologies while commercialisation has never materialised. A lack of project pipelines and an exemplary model case makes potential players even more reluctant to attempt entry into the least developed energy efficiency market in the country and the region as a whole. The foregoing challenges have been commonly and repeatedly discovered as the precedent initiatives of other emerging markets across the Southeast Asian region – Indonesia is not an exception.

COVID-19 Impacts on Energy Efficiency Promotion in Indonesia. In harsh reality, being confronted by the state budget's deficit widened due to the handling of the pandemic, the government has a strong willingness to address priority sectors such as healthcare around the procurement of vaccination and medical equipment, social protection and basic needs support that directly affect the vulnerable group of people and businesses. The governmental deployment of stimulus measures in the neediest areas seems likely to bring energy efficiency and conservation back in terms of priority. The area of energy efficiency and conservation is generally positioned as a second priority and the COVID-19 crisis has added an extra level of pressure on top of the backdrop, according to Energy Efficiency 2020 report (IEA, 2020). Given the further setbacks from the COVID-19 woes, international-level intervention in the urgent business area – lower priority though – is more important than ever: breaking a trajectory in respect of the importance of low priorities.

Designed with Contextual Lessons Learnt. The proposed Programme is designed based on the lessons learned from the precedents in the country and in other jurisdictions, in line with the strong willingness of the Indonesian government to strengthen and refine the regulatory framework and policies for promoting industrial energy efficiency in consideration of synergy impacts alongside diverse relevant programmes and initiatives.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Three Pillar Barriers-based Entry Points for Energy Efficiency Achievement. The Programme will serve Indonesia and its neighbouring countries to shift the development pathway towards a low-emission direction as prescribed in the Theory of Change (ToC) diagram. The Indonesian energy efficiency financing initiative shall address three chronic barriers – financial, regulatory, and demand-side barriers – that have been blocking the success of the energy efficiency business in Indonesia. Amid interlinked challenges and vicious circles of implementation failures, entry point initiatives as illustrated below will bring changes to the abovementioned barriers. The programme has been designed with the three barriers-based intervention in a strategic way as shown in figure 6 (Theory of Change Diagram).

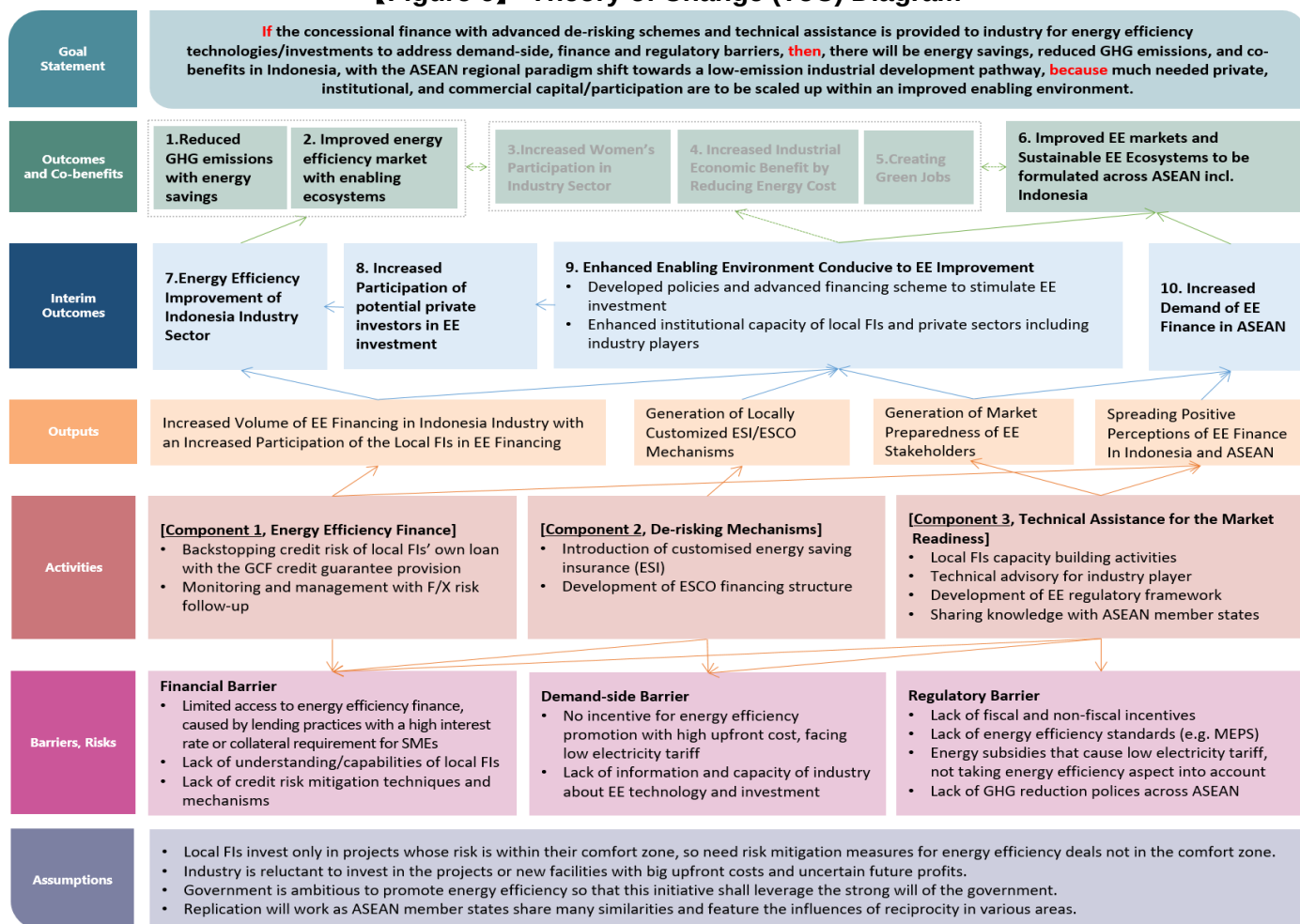
- Financial Barrier-wise Entry Point: Energy efficiency businesses will benefit from a guarantee instrument, coupled with a locally customised de-risking mechanism and capacity building and knowledge sharing activities with local FIs.
- Regulatory Barrier-wise Entry Point: Awareness and perception on detailed energy efficiency related policy or regulation will be raised by workshops and seminars for the related ministries with the support of advanced institutions with global expertise, leading into unified policy direction of the government and relevant ministries. Relevant stakeholders will be incentivised alongside active policy intervention and improved regulatory framework, including a strong need of reasonable tariff adjustments, which will

encourage the market creation and development with the deployment of energy efficient technologies in Indonesia.

- **Demand Barrier-wise Entry Point:** Local industry owners will be enhanced with mandatory capacity building and knowledge sharing activities through technical assistance (TA). Along with the TA activities, their demand for energy efficiency projects will be found through increased access to financial support such as concessional loan and guarantee instruments.

The three chronic barriers will be addressed by customised activities of individual components: financial barriers under Component 1; demand-side barriers under Components 2 and 3; and regulatory barriers under Component 3. The programme's strategic intervention will bring about an increased volume of energy efficiency finance across the country's industrial areas with proactive participation of the local FIs under Component 1, the generation of locally customised de-risking mechanisms under Component 2, and the market readiness and a spread of the best practice across ASEAN under Component 3. Such change will accelerate desirable outcomes and co-benefits that put Indonesia in the right track towards a low-carbon development pathway through reduced GHG emissions, improved energy efficiency ecosystems including the relevant market creation in and beyond Indonesia, and women's contribution to the labour market (inclusive development) and an economic leap. The programme will eventually prove that the sophisticatedly designed intervention is correct.

[Figure 6] Theory of Change (ToC) Diagram



A Pursuit of a Genuine Change through the Correct Intervention. The programme is based on the assumption that the correct intervention of addressing existing chronic barriers will empower Indonesia to

self-combat climate change through energy efficiency business development. Studies on similar precedents, extensive research, interviews, and focus group discussions with multiple key stakeholders relevant in the industrial energy efficiency ecosystem have informed on the following issues and challenges that are to be tackled under the Programme:

- Financing institutions have a limited capacity to understand energy efficiency projects and assess risks associated with the relevant technologies; thereby being highly risk-averse and reluctant to provide attractive financing terms and conditions for those who require the capital-intensive upfront cost for high efficiency technologies. Primarily motivated by the energy saving potential to improve profitability in management, SMEs are relatively more open to energy efficiency projects, and are willing to speak with banks for financing, whereas their creditworthiness and small project scale tend to discourage financial institutions – in particular, large banks and formal financial institutions – from providing finance with competitive terms and conditions. Furthermore, for SMEs, access to finance is becoming more difficult as there is currently no protection technique, such as insurance or guarantees that can mitigate the credit risk of energy efficiency projects. This situation often leads to high interest rates and rigid collateral requirements from financial institutions. Therefore, SMEs are particularly burdened with new equipment purchases or instalments requiring significant upfront costs.
- In that industries basically pursue cost reduction, comparatively higher interest costs and low energy tariffs for energy efficiency projects consequently has lowered the industry demands and has made the success of energy efficiency trials difficult over the past period in Indonesia. Also, local industry owners have not been able to make well-informed investment decisions due to a lack of knowledge about the needs and a lack of awareness of how to measure and assess the energy saving potential along with the relevant cost-benefit analysis, thereby seeing no point for replacing a still functioning technology.¹² Despite the existing regulation that commands industries whose annual energy consumption exceeds 6,000 TOE to have energy managers with periodic reporting,¹³ the relevant in-house technicians are not well trained and the reports are rarely followed up on. The main reason for this reality is due to a lack of emphasis on the importance of energy efficiency and allocated budget for clean energy, low carbon technologies R&D, and innovation, especially for developing smart manufacturing technologies. Hence, adversely affecting workforce training and education. It is identified through market assessments that Indonesia needs to overcome these barriers so as to generate demands from the industry sector.
- Ministries and public agencies have their own – varying and different – insights on the complexity of various challenges, while simultaneously acknowledging the necessity of improving and sophisticating existing energy efficiency policies and regulations with a specific focus on industry. As a result, regulatory barriers, including the lack of a regulatory framework, such as fiscal and non-fiscal incentives and low energy tariffs due to subsidies policies for fossil fuels exist, despite the Indonesian government's eagerness to respond to climate change and reduce GHG emissions. With that being said, there should be policy and regulatory synchronisation and harmonisation amongst themselves, along with an enhanced awareness of it. In order to ensure implementation on the ground, the more active interventions shall be performed through TA activities of Component 3.
- ASEAN is a political, economic and cultural community established in 1967 with an aim of pursuing peace, stability, and economic growth in Southeast Asia and promoting socio-cultural development. ASEAN member states are heavily influenced by each other in terms of economy, industry, and environment. Due to the nature of the region being greatly affected by each other, the lack of interest in

¹² GIZ and MEMR. 2019. Usulan Skema insentif & Kegiatan untuk dana energi terbarukan dan konservasi energi Fokus: Efisiensi Energi

¹³ Republic of Indonesia. 2009. Government Regulation Number 70 Year 2009 on Energy Conservation (PP No. 70/2009 KONSERVASI ENERGI)

energy efficiency projects has weakened the possibility of success and scale up of small-scale energy efficiency projects in Indonesia. Accordingly, the Programme aims to enhance awareness and regional scale up in the ASEAN region through successful pilot project implementation and sharing of lessons learned through the Programme under Component 3. This is expected to make the Programme more sustainable by the establishment of an energy efficiency ecosystem in the ASEAN region.

The Programme considers the formulation and facilitation of a self-functioning market, eventually moving towards a genuine transition to a low-carbon development pathway in Indonesia. An enabling environment is essential for creating and facilitating a market that properly functions, with Indonesia's confronting of a harsh reality that potential players and stakeholders have been barely active under the surface, getting stuck on the aforementioned stumbling blocks and obstacles. In that multiple challenges and barriers are related to each other, the Programme is designed with a set of financing and TA activities in an interlinked and supplementary manner: (1) financing energy efficiency businesses as per eligibility criteria; (2) developing de-risking mechanisms with energy saving insurance (ESI) schemes coupled with a standard performance contract and measurement and verification (M&V) tools; (3) and building capacity for market acceptance and finetuning the policy and regulatory framework, with the goal of regional scaleup and replication in mind.

Expected outputs from the Programme implementation would serve to bring about a transformative change in the industrial energy efficiency market of Indonesia and its neighbouring countries, in terms of investment and financing volume, beneficiary scale such as the number of end beneficiaries, proliferation of diverse technologies and relevant industries, quality of the policy intervention, change in perception, behavioural change, and so on. Above all, with an improved regulatory framework as a result of active policy intervention alongside training, knowledge sharing, and outreaching activities, the market stakeholders across industries will be incentivised to voluntarily choose to engage in energy efficiency and saving activities. Also, the behavioural change of the relevant stakeholders will contribute to successful track records in the country, in addition to the trust building process amongst key business stakeholders in the market. With the improvement of market acceptance and regulatory capacity, trust over energy efficient technologies and related business models will likely be strengthened and industries and financiers will make well-informed investment and lending decisions with prioritisation of energy efficient technology deployment. These changes will drive the industries of Indonesia and its Southeast Asian neighbours to be aligned with the fund-level impact for low emission development pathways as a whole at last.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
1.Reduced GHG emissions with energy savings	☐	☐	☒	☐	☐	☐	☐	☐
2. Improved energy efficiency market with enabling ecosystems	☐	☐	☒	☐	☐	☐	☐	☐

6. Improved energy efficiency market and sustainable energy efficiency ecosystem across ASEAN	☐	☐	☒	☐	☐	☐	☐	☐
7. Energy efficiency improvement of Indonesia Industry Sector	☐	☐	☒	☐	☐	☐	☐	☐
8. Increased participation of potential private investors in energy efficiency investment	☐	☐	☒	☐	☐	☐	☐	☐
9. Enhanced enabling environment conducive to energy efficiency Improvement	☐	☐	☒	☐	☐	☐	☐	☐
10. Increased demand of energy efficiency finance in ASEAN	☐	☐	☒	☐	☐	☐	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
3. Increased Women's Participation in Industry Sector	☐	☐	☐	☒	☐	☐
4. Increased Industrial Economic Benefit by Reducing Energy Cost	☐	☐	☒	☐	☐	☐
5. Creating Green Jobs	☐	☐	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The Programme consists of three components designed as an integrated package that aims to address financial, regulatory, and demand-side barriers, accelerating the facilitation of genuine paradigm shifting outcomes. Component 1 is the major private sector energy efficiency financing element and Components 2 and 3 are the technical assistance (TA) element. The TA components are designed to support a successful implementation of Component 1, which would lead to a genuine paradigm shift in the industrial energy efficiency sector of Indonesia, as well as those of neighbouring states beyond a single country.

Component (Final Beneficiaries)	Activities	Executing Entity
[Component 1] Energy Efficiency Finance (Industry Owner, Technology Provider, ESCOs)	Backstopping credit risk of local FI's own loan with the GCF credit guarantee provision	Local FIs with respect that FI's Own Loan only (PT SMI, PT IIF, Hana Indonesia IBK Indonesia, KB Indonesia, Shinhan Indonesia, Woori Indonesia)
[Component 2] De-risking mechanisms (Financial Institutions, Industry Players, Government Authorities)	Activity 2.1: Introduction of customised Energy Saving Insurance (ESI) - Development of customized insurance covering energy saving, standards contracts, energy savings validation methodologies - Consulting and in-depth interview with key stakeholder and Workshops Activity 2.2: Development of ESCO financing structure - Development of locally-fitting outcome-fixed contract - Capacity building and knowledge sharing workshops for advanced ESCO finance cases	ACE (BASE, KEA as Implementing Entities)
[Component 3] Technical Assistance (TA) for the Market Readiness (Financial Institutions, Industry Players, Government Authorities)	Activity 3.1: Local FIs capacity building activities - Support for the development of project pipeline - Technical advisory on M&E on projects Activity 3.2: Technical advisory for industry player - Technical advisory on technology installation & utilization - Technical advisory on M&E on projects Activity 3.3: Development of EE regulatory framework - Workshops and in-depth interview with policy makers - Regulatory framework for recommendations Activity 3.4: Sharing knowledge and understanding gained by the programme with ASEAN member states - Draw the lesson learnt from programme implementation - Knowledge sharing session with key stakeholders in ASEAN	ACE (Korean Banks including KDB as Implementing Entities)

Component 1 – Energy Efficiency Finance; Backstopping credit risk of local FI's own loan with GCF credit guarantee provision. Component 1 shall establish the guarantee facility that offers appropriate de-risking mechanisms that are currently lacking in the market for successful energy efficiency trials of local FIs. The Programme provides GCF credit guarantee to LFIs (guarantee beneficiary) through KDB, backstopping credit risk of relevant financial beneficiaries as borrowers of loan to be provided by LFIs.

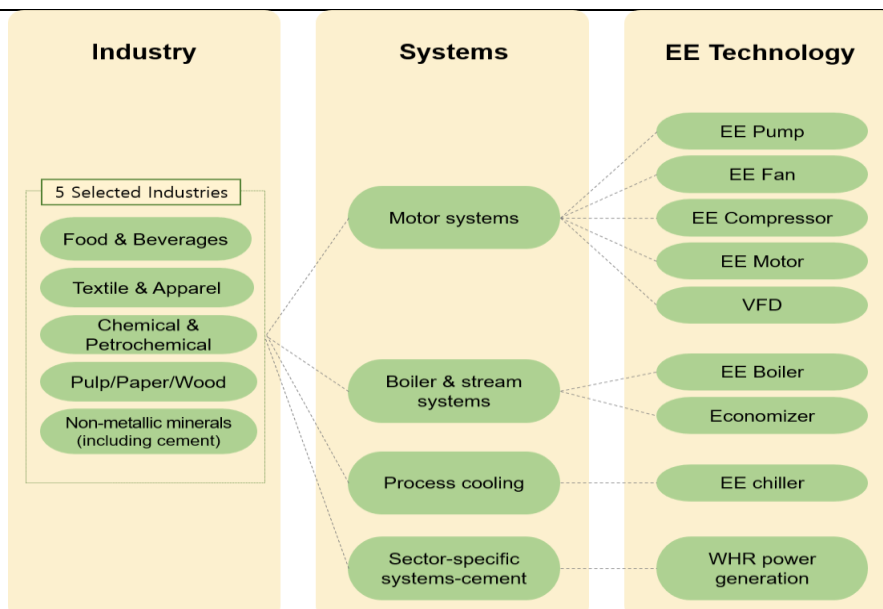
• **Target Selection for Priority Industries and Energy Efficient Technologies.** The Programme identifies five priority areas for energy efficiency financing implementation from the industrial sector, as a result of the Market Assessment. The industrial subsectors – (1) food and beverage, (2) textiles and apparel, (3) pulp and paper and wood, (4) chemical and petrochemicals, and (5) non-metallic minerals – are selected with the quantitative criteria of Table 4 below .

【Table 4】 Criteria to Identify Priority Subsectors

Energy use and GHG emissions-related criteria
• Subsector energy use share/ranking • Subsector GHG emissions share/ranking • Electricity vs fuel use in the subsector
Systems and Technology-related criteria
• Importance of our targeted systems (e.g., motor systems and boiler systems) in the subsector • Share of targeted systems' energy use from total subsector energy use • The potential application of targeted technologies (e.g., efficient pump, fan, variable frequency drive (VFD), boiler, etc. in the subsector)
Economic-related criteria
• Share/rank of subsector from total industrial value-added/GDP • Share/rank of subsector from total industrial export/import • Strategic importance of subsector in Indonesia
Other criteria
• Overall financial readiness of the subsector • Subsector's reputation in willingness to change and invest in energy efficiency • Small and medium enterprises (SMEs) vs large energy-intensive subsector • Existing regulations in Indonesia and how it impacts each subsector

The five industrial subsectors, identified through a first-round analytic selection, were further narrowed down by a second-round analytic selection that included a review of industrial, systems, and technology levels. The industrial level criteria included factors such as energy use, GHG emissions and economic contribution, for example, and the systems level criteria embraced the cross-cutting nature of systems for market transformation and the importance in the relevant industry. The technology level criteria contained energy efficiency and mitigation potential, cost-effectiveness, and local capacity to install and operate the relevant technologies. Based on the three levels of criteria, the Programme selected nine industrial energy efficiency technologies: pump, fan, compressor, variable speed drive (VSD), motor, steam boiler, economizer for boilers, chiller, and waste heat recovery (WHR) for power generation in cement plants. For more details, please see the Market Assessment submitted as Annex 2.

【Figure 7】 Selected Industrial Subsectors with Rankings by Criteria



• **Eligibility Criteria and Terms and Conditions (T/C) Customised for Indonesian Energy Efficiency Projects.** Partner local FIs are to provide energy efficiency finance to ultimate end beneficiaries pursuant to pre-defined eligibility criteria summarised in Table 5 below. The criteria identify priority target industries through consideration of energy-intensive areas, priority target systems and technologies, qualification of borrowers, and other quantitative and qualitative sources including market assessments and focus group discussions. For instance, the NDA of Indonesia emphasises the needs of vulnerable locations and beneficiaries as prioritised in the National Medium Term Development Plan (RPJMN) 2020-2024 and in turn, the Programme assigns priority to borrowers who belong to the identified group. The selection criteria are treated to function as a protective set of guidelines that supports the GCF and KDB standards: e.g., a gender investing lens is applied to the Programme by prescribing a high priority on women-led SMEs in terms of access to finance. The criteria are expected to be specified further during the implementation stage according to more in-depth practical and legal discussions with local FIs. In addition, the criteria for the selection of beneficiaries under Components 2 and 3 are set out in the Term Sheet (Annex 14).

【Table 5】 Eligibility Criteria for Energy Efficiency Financing

Final Beneficiaries (Borrowers)	• <u>General – Mandatory Qualification</u> Industry owners, enterprises, technology providers, ESCOs, or project developers in any form of a greenfield, replacement, upgrade, scale-up and/or expansion, which aim to enhance energy efficiency in industrial areas of Indonesia. * Entities with more than USD200million in annual sales not eligible * Guarantee coverage to a single borrower shall be up to USD 20 million (20% of total GCF guarantee cover).. • <u>Flexibility in Form</u> A single final beneficiary may be selected on a stand-alone basis or on an aggregated basis (e.g., bundling a number of projects with a mixture of different energy efficiency technologies) or multiple final beneficiaries may be grouped into a single Sub-project Loan Agreement (e.g., bundling a number of industry owners with their respective projects within an industrial complex). • <u>Priority Condition 1 (Indonesia NDA priority)</u> Vulnerable locations and beneficiaries as prioritised in the National Medium Term Development Plan (RPJMN) 2020-2024, NDC roadmap, and NAPs. • <u>Priority Condition 2 (Women's Empowerment and Gender Investing Strategy)</u>
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	A women-led SME will be selected with priority. ¹⁴ Even if not gender-mainstreamed yet, a company with a commitment to gender mainstreaming could be prioritised under the guidance of gender experts employed for the Programme.									
Total Cost	- General – 100% of total amounts - The cost of purchasing the energy efficient technologies and equipment; - Required associated costs/expenses for the energy efficient technologies and equipment, including freight, installation, customs duty, and related insurance; and - Other costs/expenses considered necessary for the project implementation by relevant FIs.									
Equity Injection Requirement	- As per the classification of the final beneficiaries into Class A / Class B, a minimum level of equity injection is required as 5% / 10% of Total Cost, respectively. - Each of the final beneficiaries is classified as per the annual sales volume, based on SMEs classification of WEF(World Economic Forum), in 2021¹⁵ (USD converted) as follows:<table><tr><th>Annual Sales Volume</th><th>Classification</th><th>Equity Requirement</th></tr><tr><td>Up to USD35million</td><td>Class A/ ESCOs</td><td>5% (of Total Cost)</td></tr><tr><td>USD35million ~ USD200million</td><td>Class B</td><td>10% (of Total Cost)</td></tr></table>	Annual Sales Volume	Classification	Equity Requirement	Up to USD35million	Class A/ ESCOs	5% (of Total Cost)	USD35million ~ USD200million	Class B	10% (of Total Cost)
Annual Sales Volume	Classification	Equity Requirement								
Up to USD35million	Class A/ ESCOs	5% (of Total Cost)								
USD35million ~ USD200million	Class B	10% (of Total Cost)								
Total Eligible Investment Cost	Total Cost excluding the equity injection by final beneficiary									
Environmental and Social Safeguards (ESS)	- To be classified as “E&S Risk Category B” or “E&S Risk Category C” - The Programme will not finance any project with the category A potential - The Programme will not finance any project with potential adverse impacts in relation to indigenous peoples in Indonesia - The Programme will not finance any project that may involve land acquisition that may cause any involuntary resettlement or physical/economic losses - The Programme will not finance any project with potential adverse impacts on ecological diversity and ecosystem - The Programme will not finance any project with potential adverse impacts on cultural heritage									

¹⁴ IFC’s definition of women-owned enterprise i.e., at least 51% owned by a woman or women; or a business where at least 20% is owned by women with one or more women in a top management role (CEO/COO), and at least 30% of the board of directors being female (where a board exists). But to better suit the objective to this programme, broader definition of women-owned business by MCA-Indonesia (2016) will be adopted, which is business in which a woman or group of women holds most of the important roles and positions in the ownership of capital/business inputs, business management, human resources/labor, and/or business control. Consequently, a thorough assessment should be carried out in a case-by-case basis to determine whether a business can be categorized as a women-owned business.

¹⁵ www.weforum.org/agenda/2021/09/how-can-indonesian-sems-scale-up/

Priority Industries & Technologies

- General – Mandatory Qualification
 - Technologies with energy efficiency improvement by more than 20%, compared to the existing equipment of a beneficiary; for cases where 20% is not achievable (e.g: motors), best in class equipment will be used
 - The rate of improvement would be proven by evidence, such as catalogue/brochure/declaration from suppliers, reports/certificates from certified energy auditors/independent advisors, or reports developed under TA activities of Component 3;
 - No re-sale of existing equipment is allowed; and
 - Second-hand equipment is not eligible
- Priority Industries
Food and beverage, textile and apparel, chemical and petrochemical, pulp / paper / wood, and non-metallic minerals (including cement).
- Priority Systems and Technologies
 - Pump, fan, motor, compressor within the motor system (e.g., high efficiency pump, fan, compressor, motor, variable speed drive (VSD) and variable frequency drive (VFD) etc.
 - High efficiency boiler / economiser / chillers within boiler / steam / process cooling systems
 - Sector-specific system (e.g., waste heat recovery (WHR) power facility in cement industry)

A credit guarantee coverage ratio ranging between 50 to 95% will be offered based on the market conditions at the time of the establishment of the FI own loan and guarantee agreement, the FI's risk appetite, and the size of the final beneficiary. The guarantee coverage will provide an adequate incentive for local FIs to provide their own loans for those who have greater needs for financing support for their projects, such as the less creditworthy SMEs, who typically have substantial difficulty obtaining loans with competitive rates. The credit guarantee instrument would function in a way that ensures the balance between comfort against credit concerns and risk-sharing amongst parties. In addition, there is a substantial need for support for energy service companies (ESCOs) according to the Market Assessment. Even though ESCOs have already operated in Indonesia for a long time,¹⁶ their impact on the energy conservation sector is considered minimal, as Indonesian ESCOs struggle to obtain third-party finance due to their weak balance sheets and lack of collateral. Therefore, for the Programme, ESCOs are categorised in the same group as less creditworthy SMEs. Component 1 will command at least 50% of loans provision for the vulnerable SMEs / ESCOs.

【Table 6】 Guarantee Coverage Ratio

Annual Sales Volume	Classification	Coverage Ratio
Up to USD 35 million	Class A/ESCOs	Up to 95% (of Total Eligible Investment Cost)
USD 35 million ~ USD 200 million	Class B	Up to 50% (of Total Eligible Investment Cost)

• **Selection of Local Financial Intermediaries with Experiences in the Local and Global Energy Efficiency Financial Market.** The Programme has considered a strategic partnership with local FIs for the proper development, implementation, and management of the Programme, leaving primary ownership with local stakeholders. The FI selection process has considered multiple factors: (1) extensive commercial lending experiences in industries including energy-intensive industries; (2) familiarity and/or understanding of the guarantee structure through collaboration with the Multilateral Development Banks (MDBs), Development Financial Institutions (DFIs), and/or global financiers; (3) familiarity with and/or willingness to

¹⁶ In 1982 already, PT Koneba (now known as PT.EMI) was established to serve as a company focused on energy conservation and energy management (ACE, 2017). In 2018, the ESCO Association APKENINDO reported 25 members (Carbon Trust 2018). Research in 2017 found that ESCOs in Indonesia are usually small engineering companies that provide industry and buildings services

adopt global standards through the Programme implementation; for example, in terms of environmental and social (E&S) safeguards, gender mainstreaming policy and system, procurement, and ethics and compliance; and (4) strong willingness to be a pioneer for climate finance transactions.

Through a series of extensive market research, discussions, and interviews with potential FIs in Indonesia during the Market Assessment phase, several local financial institutions have been confirmed as partner FIs and Executing Entities under Component 1 to enjoy the merits of the Programme. In particular, KDB has selected FIs in two groups, which are 1) “LFIs with local experiences” with a wide local network, broad experiences, and an understanding of the local government policy and 2) “LFIs with global experiences” that can transfer experiences and knowledge from advanced energy efficiency projects and climate financing in Korea. Other FIs are to be further selected to expand this opportunity to include as many FIs as possible. The invitation and selection of new FIs and associated capacity assessments will be completed by KDB prior to the FAA effective date.

1) LFIs with local experiences

The Programme provides opportunities to further develop synergies with existing local players, such as PT SMI (SMI) and PT IIF (IIF). The institutions are committed to contributing to sustainable development, and as infrastructure financing companies, they are mandated to provide innovative financing schemes to infrastructure sectors. As they have expanded their business sectors, renewable energy and energy efficiency have become priority areas of focus, allowing them to gain valuable insights with financing energy efficiency projects as they take on more projects to garner experience in the sector. As relatively new players to the market, they, and other interested institutions with a lack of EE project experience, may encounter unfamiliarity with the risk profile of energy efficiency financing. The de-risking mechanism can serve to overcome the main barriers for financial actors to invest in energy efficiency, and they can play a key role in creating an innovative financing mechanism for energy efficiency in Indonesia.

In addition, both institutions have been closely working for GCF-related businesses – SMI is a GCF accredited entity and has been in the process of developing GCF projects, and IIF is in the process of becoming accredited by the GCF. The participation of these two local banks in this Programme will strengthen their capacity for global climate project development, contributing to further expansion of the GCF Programme and its vision in Indonesia and the neighbouring ASEAN region.

2) LFIs with global experiences

As part of Component 1, KDB, as the Accredited Entity, with the Korea Energy Agency will lead a consortium of leading financial institutions from Korea for the Programme to pass on key lessons learnt from energy efficiency financing experiences within Korea and internationally, including in Indonesia. KB Kookmin Bank, Hana Bank, and Industrial Bank of Korea have committed to their participation in the Programme, both in Korea to support at the headquarters level and locally with their domestic branches in Indonesia. Other companies with existing business understanding and physical presence and experience in Indonesia were reached out to participate in the Programme, and currently, Woori Bank and Shinhan Bank have also expressed their strong interests and intentions to participate. Interviews were conducted with Korean companies (from electronics and technology service industries) with prior experiences with energy efficiency projects within Indonesia and valuable feedback was shared, from recommendations on financial structures of financing and capacity building for local industry stakeholders to what to watch out for in terms of risks such as higher fees in lieu of lower interest rates and experiences in energy saving facility installation projects through the ESCO

mechanism. The Programme will continue to gather useful insights and provide those experiences to the executing entities of the Programme to leverage and learn from.

【Table 7】 LFI with Global Experiences

Consortium led by KDB

Accredited Entity



A state-owned bank committed to promoting and disseminating the concept of sustainable development and its promotional activities

Local FIs: Indonesian branches of Korean banks



Industrial Bank of Korea

Government bank is the largest player in SMEs loan market



KB Kookmin Bank

Korea's largest commercial financier



SHINHAN BANK

Korea's first civilian-controlled financial holding company



Hana Bank

Commercial bank with impressive global network



WOORI BANK

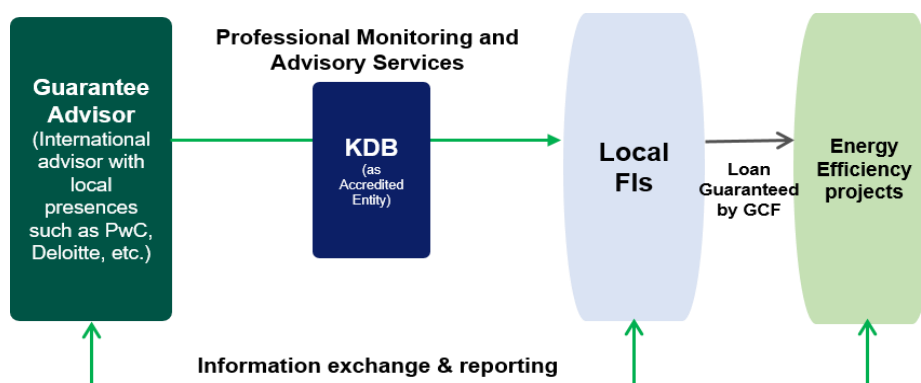
Established as Korean national bank specialized in ASEAN market

• **Monitoring in line with Local Financial Market Needs.** Local FIs, as well as government stakeholders, highlighted throughout the interviews with the AE (KDB) and the Executing Entity of Components 2 and 3 (ACE) that the F/X rate risk can be a main barrier for successful Programme implementation given the nature of Indonesia's business environment requiring FI own loans to final beneficiaries to be in Indonesian local currency and the GCF Credit Guarantee is only available in USD. KDB has developed the GCF Credit Guarantee structure in line with these market needs, designing the guarantee so that the GCF will partially absorb the foregoing F/X risk and the rest of F/X risk is borne by FIs, thereby reducing one of the major concerns of potential FIs for the Programme and simultaneously enhancing competitiveness of the ultimate interest rates for final beneficiaries on the ground. Focusing on the fact that F/X risk will increase as the loan period increases, F/X risk absorption by GCF is structured in Annex 14. This risk will be monitored and immediately reported to GCF by KDB based on the Guarantee Advisory and KDB's experiences in the international financial market

In addition, KDB, the Accredited Entity, plans to hire a global independent expert with a local presence in Indonesia as a Guarantee Advisor, who will provide professional monitoring services for each sub-project, related performance, and any change in credit risk with a full review of related contracts including sub-project loan and Guarantee Agreements. The monitoring activity will be immediately reported to KDB to support its close Programme monitoring, and also to be shared, if considered necessary by the AE, with LFIs to assist in their successful Programme implementation.

This structure including Guarantee Advisory is developed through interviews with various local stakeholders. For the sake of clarity, the selected Guarantee Advisor will be hired not using GCF funding but through the AE fee, under KDB's willingness to expand its contribution to the Programme.

【Figure 8】 Guarantee Advisor

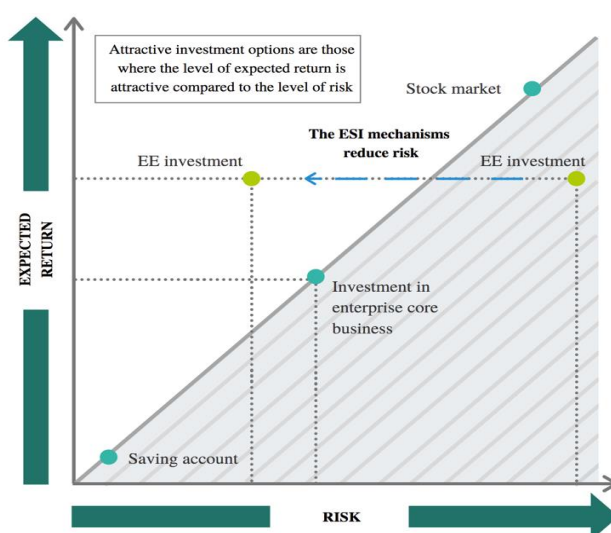


Component 2 – De-risking mechanisms. Component 2 introduces de-risking mechanisms of an ESI / ESCO model and scheme that helps lower the perceived high level of risk and establishes credibility and comfort amongst key relevant stakeholders. To avoid any confusion, this is not a financing component to provide any de-risking scheme to each project but a Technical Assistance (TA) component to introduce and develop de-risking mechanisms for better energy efficiency projects for the future in Indonesia.

Activity 2.1. Introduction of customised Energy Saving Insurance (ESI). Key stakeholders including local FIs will be convened and consulted on the devised de-risking mechanisms to ensure buy-in. During the Programme development phase, KDB has discussed the ESI model with BASE, a Switzerland-based non-profit foundation. According to BASE, “for investors, the risk-return trade-off is one of the essential components of each investment decision” and the critical issue is that “energy efficiency projects face a high level of perceived risk and the expected reward is not commensurate with the perceived risks.” Activity 2.1 shall therefore propose the ESI model that addresses such challenges by reducing the investment risk of enterprises and building trust between relevant stakeholders, which will lead to a more attractive risk-return trade-off perception in the market. For this, BASE, with great expertise in the de-risking model, will be considered as an Implementing Entity, which is a service provider engaged by ACE (Executing Entity) for the implementation of this activity.

The Programme plans to introduce the ESI scheme to the Indonesian energy efficiency market using multiple examples from advanced countries. Moreover, it is expected to develop a customised ESI contract and validation methodologies through workshops or consulting with local stakeholders including LFIs.

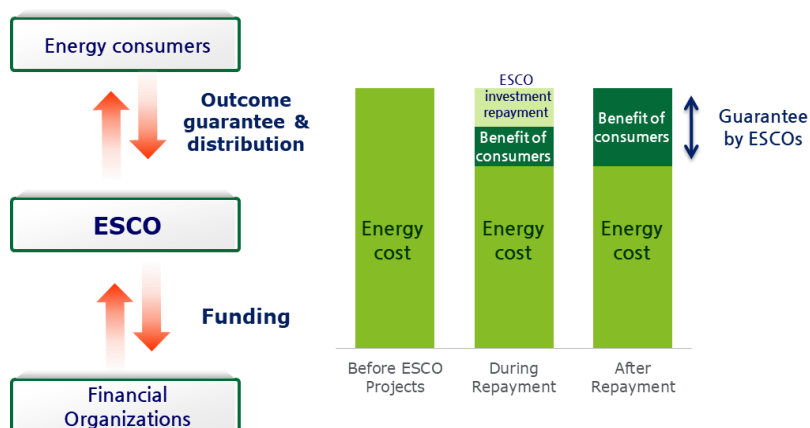
[Figure 9] Risk-return Trade-off for Enterprises



Activity 2.2. Development of ESCOs financing structure. As mentioned in Component 1, there is a substantial need of support for energy service companies (ESCOs) according to the Market Assessment.

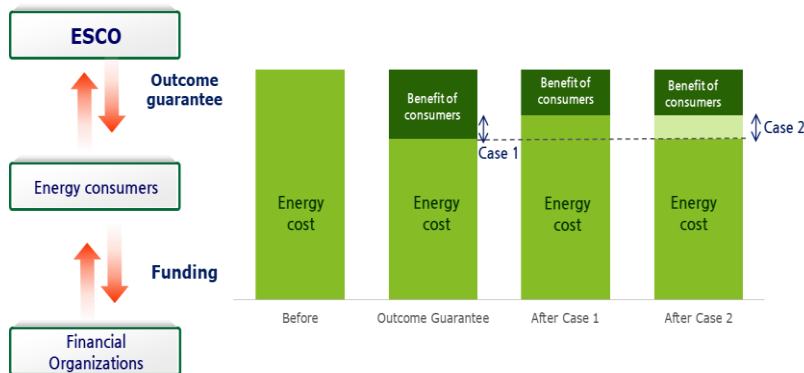
ESCOs in Indonesia are primarily small engineering companies that provide industry and building services, and they struggle to obtain third-party financing due to their weak balance sheets and lack of collateral. For better access to third-party finance, PT. EMI, a state-owned ESCO in Indonesia, suggested the business model as Figure 10, outcome-guaranteed financing by ESCO, through interviews with ACE, Executing Entity of Component 2. The outcome-guarantee financing is when ESCO withdraw the investment via saved energy amount or cost, which is the benefit of the customers, is guaranteed by ESCOs.

[Figure 10] : Outcome-guaranteed Contract (ESCO-financed)



In order to support ESCO financing, the Programme provides up to a 95% guarantee coverage ratio to ESCOs who have tried to secure third party finance, including through the ESCO financing structure above, categorising the ESCOs in the same group as less creditworthy SMEs in Component 1. In addition to guarantee funding, the Programme, through Activity 2.2., will provide advisory services for the ESCOs to achieve better financing structures and obtain knowledge sharing from advanced institutions with expertise in ESCOs financing. The Programme plans to develop a customised ESCO financing scheme and a standard contract through the Executing Entity, ACE, and knowledge sharing and workshops through the Implementing Entity (In-kind contribution), the Korea Energy Agency (KEA) , which is a service provider engaged by ACE (Executing Entity) for the implementation of this activity. KEA is Korea's energy leader and an experienced organisation for energy efficiency promotion and relevant economic development. KEA will step in to provide opportunities for ESCOs with high human capacity but low capital, which hinders their achievement of sustainable growth. The agency will build capacity and help create awareness of advanced energy efficiency financing (EEF mechanisms) to ESCOs, such as consumer outcome-guaranteed contracts. The outcome generated contracts can be financed by energy consumers unlike the previous ESCO financing scheme, which can utilize the consumers' credit that is usually better than the one of ESCO. Its detail is outlined in Figure 11 below.

[Figure 11] : Outcome-guaranteed Contract (Consumer-financed)



For the outcome-guaranteed contract, consumers are responsible for the delivery of funding. The savings amount is guaranteed by ESCOs and depending on the level of guaranteed saving amount vis-à-vis the actual saving amount, a cash transfer may be necessary. When the guaranteed savings amount is more substantial than the actual savings amount, ESCOs would pay back the difference in cash; in cases when the guaranteed savings amount is lower than the actual savings amount, the difference

distributed would be agreed upon between ESCOs and energy consumers.

Component 3 – Technical Assistance (TA) for the Market Readiness. One of the major barriers for successful energy efficiency finance is the lack of experience and understanding about energy efficiency by major stakeholders, which can be classified by consumer – industry, supplier, and financial institutions, and external environment – regulatory framework, in terms of energy efficiency finance. In Component 3, successful energy finance with balanced roles of stakeholders is pursued through capacity building for each stakeholder, which will support the Indonesian market that is ready to accept advancements in energy efficiency, in line with the GCF standards in terms of environmental and gender risk management. To avoid any confusion, this is not a financing component to provide funding to any installation of new technology, etc., but a Technical Assistance (TA) component to build the capacity of local stakeholders including financial institutions, industry players, government authorities, and neighboring countries for better energy efficiency eco-system in Indonesia and ASEAN member states. .

Activity 3.1. Local FIs capacity building activities. This activity focuses on the establishment of reliable two-way communication channels to provide continuous support to local FIs, who are EE under Component 1. ACE, as the Executing Entity for Component 3, will organise regular committee meetings and workshops to exchange knowledge and lessons learnt regarding energy efficiency financing to support in their efforts in the market and provide open communication channels between the localised Korean banks' subsidiaries and local financial institutions in Indonesia, with a support of KDB, as the Accredited Entity. In particular, the subsidiaries equipped with both local presence and global energy efficiency finance knowhow, through this committee, will shares advanced financial experiences and systems with the local banks, especially related to climate finance and ESG principles under ACE's support, which means that the localised subsidiaries will act not only as the Executing Entities in Component 1, but also as the Implementing Entities (in-kind contribution) for Activity 3.1.

Activity 3.1 also plans to establish a programme implementation unit (PIU) in local FIs, primarily SMI and IIF, for the sake of practical support in terms of pipeline development, investment portfolio management, monitoring and evaluation for sustainability, and reporting to KDB on a regular basis to ensure success in the energy efficiency finance ecosystem. The PIU will be composed of GCF project managers, focal points, and consultants dispatched from external expert agencies or TA Providers under Activity 3.1. The Programme will require LFIs to establish a PIU where local Technical experts (incl. E&S and gender) are to be assigned and to provide advisory to support the development of pipeline, technical, E&S monitoring, evaluation & assessment activities of FIs, and even the developoent of pipeline. The anticipated pipeline will be in compliance with a set of eligibility criteria agreed upon by the GCF and KDB (Table 5), with a financing roadmap and time-bound action plan, and an operational capacity of implementing the Programme in line with sustainability principles, E&S, and gender safeguards.

Activity 3.2. Technical advisory for industry player. The focus for Activity 3.2 is to support industry stakeholders, who will be the final beneficiary of the energy efficiency finance. Industry players who are expected to be selected as borrowers under Component 1 in line with Eligibility Criteria (Table 5) will be beneficiaries under Activity 3.2. The main purpose of this activity is to boost the demands of industry stakeholders regarding energy efficiency projects by informing them of the need and economic benefits of such energy efficiency programmes and to help them feel that the projects are more accessible through training and technology advisory services. The activity supports the installation and utilisation of new energy efficient technologies to final beneficiaries, through seminars and workshops for industry players. In addition, it plans to implement a successful pilot energy efficiency project by supporting the industry stakeholders in their communication with the FIs and reporting in line with GCF and KDB guidelines during programme implementation.

Activity 3.3. Development of EE regulatory framework. This activity will support Indonesia to fine tune, strengthen, and document the regulatory framework of the energy efficiency investment and technology as a result of review and a series of consultations with Indonesian policy makers: e.g., adoption of fiscal and/or non-fiscal incentives on energy efficiency actors, and the labelling and MEPS for industrial technologies.

According to the Market Assessment (Annex 2), the Indonesian government is not unaware of the need of an energy efficiency programme. However, there seems to be a lack of deep understanding on energy efficiency projects, finance, and policies and their ability to implement energy efficiency policy and regulation in detail. Since economic development is an important goal for all developing countries, including Indonesia, energy efficiency policy and regulatory frameworks are likely to be second to economic policy; therefore, subsidies for fossil fuel are prolonged. In addition to the problems in Indonesian policy including subsidies for fossil fuels, there are still big gaps of opinions and perception on energy efficiency between each ministry – such as Ministry of Environment, Ministry of Economy and Finance, and Ministry of Industry and Energy, which is one of main barriers. The Programme will narrow this gap through workshops for government departments and seminars to raise a clear awareness of the need for energy efficiency promotion to policy makers, and draw a new desirable policy direction. In this situation, the main goal of this activity is to strengthen the capabilities of government agencies and to show the potential of simultaneous development between the economy and energy efficiency through successful implementation of Component 1 and sharing the lessons learnt, which will induce the government to implement energy efficiency policies and refine the related frameworks voluntarily.

Activity 3.4. Sharing knowledge and understanding gained by the Programme with ASEAN member states. Regional replication of the Programme throughout ASEAN member states is intended for this activity, bearing in mind that not only Indonesia, but also the entire ASEAN region, has experienced tremendous increments in energy demand, and that the long-term energy supply outlook is likely less positive in the absence of energy efficiency and conservation measures. Component 3 shall collect, organise, disseminate, and display valuable lessons learnt, stimulating knowledge sharing actions for brainstorming and devising contextual fitness-based solutions for countries in needs. ACE, the Executing Entity of the component, has assessed existing energy efficiency and conservation financing schemes through the ASEAN region as below (see Table 9), which will be referred to as baseline with other considerations, when it comes to shortlisting candidate countries. In that most energy efficiency projects in this region are currently carried out through grants, including for Indonesia, lessons in energy efficiency guarantees experienced through this Programme can present a fresh financing perspective for other ASEAN countries in situations and circumstances similar to those of Indonesia.

[Table 9] Energy Efficiency & Conservation Financing Schemes in ASEAN Member States

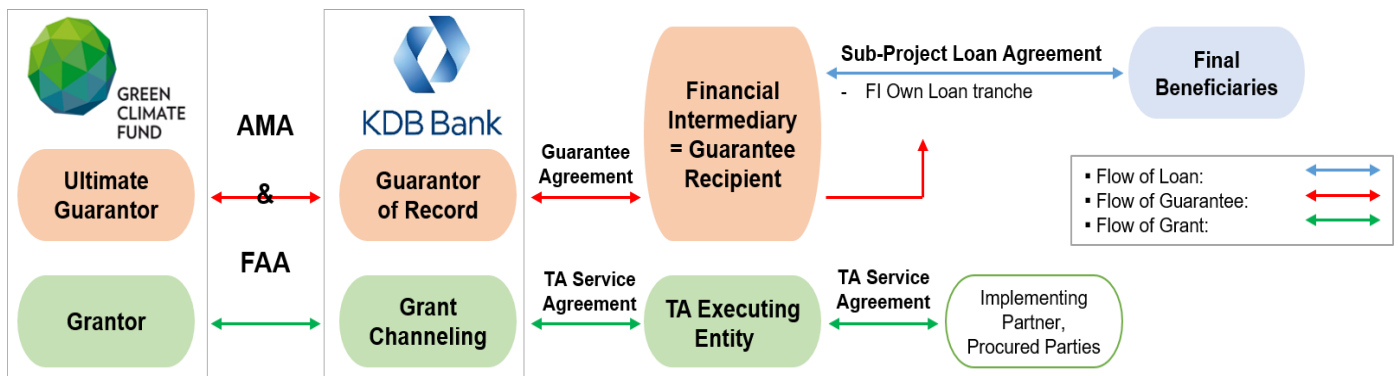
COUNTRY	DEDICATED EE GRANTS	DEDICATED EE LOANS	DEDICATED EE EQUITY	DEDICATED EE GUARANTEE	EPC
Brunei Darussalam					
Cambodia					
Indonesia	■				
Lao PDR	■				
Malaysia	■	■		■	■
Myanmar	■	■		■	
Philippines	■				
Singapore	■				■
Thailand	■	■	■	■	
Vietnam	■	■			

(Source: ACE, 2018)

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Legal and Contractual Arrangements under the Standard Two-step Financing Intermediation Programme. The Programme features a standard financing structure under the supervision of KDB, the Accredited Entity, and engagement of local FIs, ACE, and other expert groups, executing and implementing entities, to promote energy efficiency businesses. Its umbrella agreement is a Funded Activity Agreement (FAA) under which other legal and contractual arrangements are to be set and effective throughout the course of the FI own loan programme.

【Figure 12】 Legal and Contractual Arrangement with Financial Flows

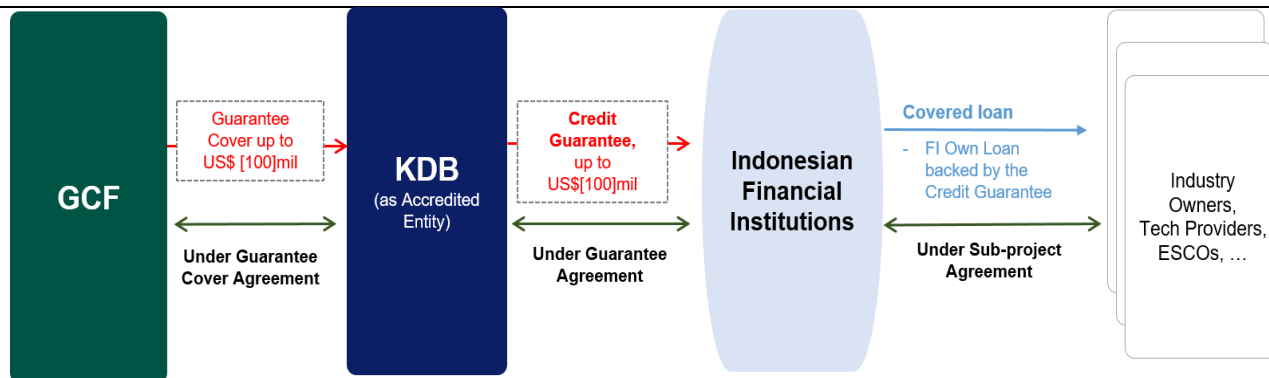


Funded Activity Agreement (FAA): In accordance with the Accreditation Master Agreement (AMA), the GCF and KDB will enter into the FAA in a form of the agreement below for each component, whereby KDB will administer the GCF proceeds and manage the Programme during the Programme lifespan.

- Component 1: Guarantee Cover Agreement
- Components 2 and 3: Grant Agreement

Under the umbrella agreement of the FAA, KDB shall pursue the following legal and contractual arrangements. Figure 13 illustrates the overall implementation structure and contractual, financial, and institutional arrangements between participants under Component 1.

【Figure 13】 Implementation Structure of Component 1



Guarantee Agreement. (Subsidiary Agreement under Component 1) A Guarantee Agreement, whose terms and conditions are set out in accordance with those in Term Sheet (Annex 14), will be entered into between KDB (guarantor of record) and each FI (guarantee beneficiary), guaranteeing the credit risk (Credit Guarantee) of the relevant final beneficiary (borrower) who will borrow covered loan from local FIs under a Sub-project Loan Agreement.

Guarantee Cover Agreement. GCF will provide guarantee cover to KDB so that it discharges its payment obligations under the Credit Guarantee given by KDB to local FI as the provider of the guarantee cover under the Guarantee Cover Agreement which will be entered into between GCF and KDB in accordance with the terms of the FAA.

Table 10 explains the details of GCF Guarantee Cover in the overall perspective: from the moment when a local FI gets entitled to draw on the Credit Guarantee upon default to how GCF's Guarantee Cover is given and flows back to the local FI. This can be read in conjunction with Table 10 – Term Sheet for GCF Guarantee Cover Agreement.

Table 10 – GCF Guarantee Cover Structure

GCF's Obligation	A. GCF pre-funds KDB on a "look forward" basis for any calls made on KDB guarantees. Claims issued by Local FIs under Credit Guarantees would constitute "eligible claims" under the Guarantee Cover agreement B. Specifically, subject to the satisfaction of relevant conditions precedent (including the issuance of a disbursement request by KDB), GCF pre-funds and pays to KDB "eligible claims" under the Credit Guarantee which are reasonably expected to arise within six (6) months from the date of the relevant request for disbursement
KDB's Obligation	A. To the extent that GCF has paid to KDB and KDB has not used the funds received from GCF to discharge claims under the GCF Guarantee within six (6) months from the date of the relevant request for disbursement, KDB may (i) continue to hold such pre-funded and unused funds ("Unused Amounts") and setoff such amounts against its next disbursement request to GCF; or (ii) reimburse GCF the Unused Amounts at zero interest, noting that such sums reimbursed shall not be available for further disbursement. B. Alternatively, to the extent that GCF has paid to KDB and KDB has used the funds received from GCF to discharge claims under the KDB guarantees within six (6) months from the date of the relevant request for disbursement

- | | |
|--|---|
| | <p>a) KDB must pursue the underlying borrowers and provide recoveries back to GCF, and ensure that GCF is able to be provided with subrogation rights to independently pursue the borrowers; and</p> <p>b) if KDB fails to recover from the underlying borrowers within an agreed period of time, KDB must transfer its subrogation rights to GCF.</p> <p>C. KDB must pay to GCF the commercially agreed portion of the guarantee premium that KDB receives from Local FIs, as a “guarantee cover fee”.</p> |
|--|---|

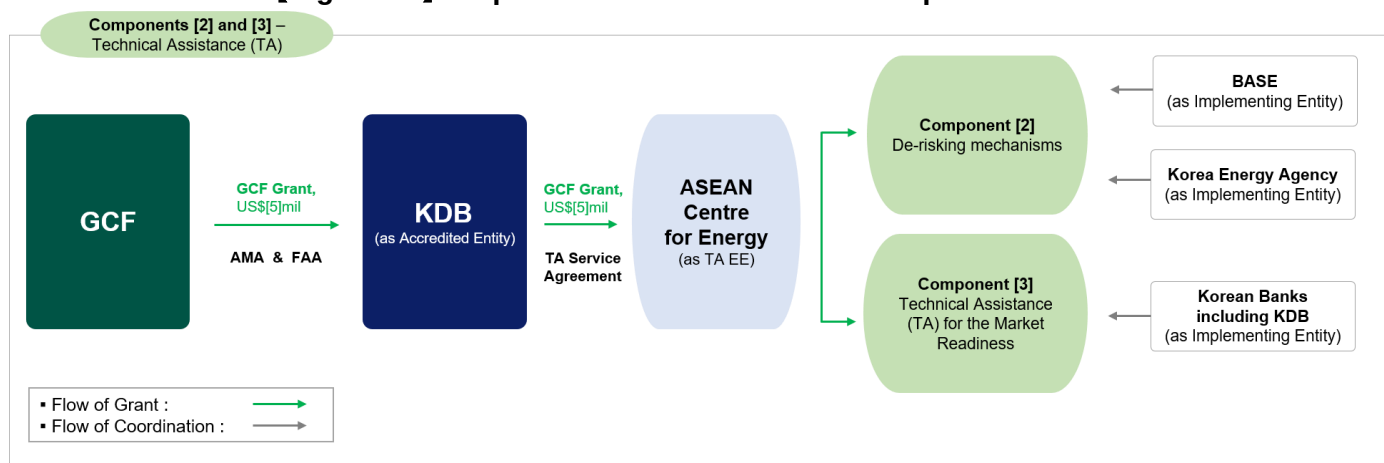
Each of the FIs, as an Executing Entity, will be responsible for ensuring that each Sub-project is in line with the Eligibility Criteria set out in B.3 and, all necessary or desirable due diligence has been conducted. The Accredited Entity will be responsible for the overall management, implementation and supervision of each FI regarding the relevant Guarantee Agreement and related Covered Loan in line with the Accredited Entity's own internal rules, policies and procedures pursuant to the AMA.

Sub-project Loan Agreement. A Sub-project Loan Agreement is to be contracted between a local FI and each respective end borrower, which implies a number of Sub-project Loan Agreements to be contracted on the ground. As per the financial instrument needs of end borrowers, terms and conditions of the agreement would be customised with credit lines available under the FI own loan backed by the Guarantee Agreement.

TA Service Agreement. (Subsidiary Agreement under Component 2 and 3) KDB and ACE, an Executing Entity of TA components (Components 2 and 3), will enter into a TA Service Agreement, aligned with the FAA. Pursuant to the TA Service Agreement, ACE will be responsible for TA activities of the Programme, in partnership with implementing entities with relevant expertise: e.g., BASE is well-versed in energy saving insurance modelling. ACE will be guided by KDB for selecting proper Implementing Entities and utilising the GCF grant proceeds in the most effective way.

The overall implementation structure and contractual, financial, and institutional arrangements between participants under the Components 2 and 3 are illustrated in Figure 15 below:

【Figure 15】 Implementation Structure of Components 2 and 3



A Proven Capacity of Engaging Entities to Deliver Aimed Results. Relevant expertise and experiences of individual entities – accredited entity, executing entity, implementing entity, and partners in various forms – are described in Table 10 below. The overall implementation structure and contractual, financial, and

institutional arrangements between participants under the Programme are illustrated in Figure 15 as shown above.

【Table 10】 Role and Capacity Assessment of Engaging Entities

KDB (Accredited Entity)	• GCF Proceeds Channelling; • Follow-up management of GCF Guarantee Cover under Component 1; and • Monitoring, Supervision and Evaluation of all the executing entities' implementation under all Components • E&S and gender mainstreamed system well institutionalised • Equipped with a stable credit rating based on strong macroeconomic fundamentals of Korea with having successfully completed financial arrangements across countries; e.g., top-tier PF arranger in Asia Pacific; • A proven track record of deploying energy efficiency finance while creating energy efficiency markets on scale and tackling challenging circumstances of the then least developed market in Korea; * KDB has over 20 years of experience in domestic energy efficiency programme and has a loan balance of more than USD 300 million as of the end of 2020. • On-lending is one of many business areas of the bank. As a policy financing institution, KDB has possessed a proven track record of two-step lending experiences under a structured internal system with intermediaries. • Willing to transfer decades-long accumulated experience and know-how on taking on calculated risks and strategically creating financial values in new business areas with a purpose of the ASEAN regional replication and scale up; and • Well-versed in a standard on-lending scheme with supporting multiple sub-projects and sub-borrowers in collaboration with local FIs in Southeast Asia.
PT IIF (Executing Entity for Component 1)	• Local Financial Intermediary under Component 1; and • Monitoring, Supervision and Evaluation of the Final Beneficiaries' implementation under Component 1. • A strong track record of being engaged in infrastructure financing including Renewable energy plant and advisory services that are professionally managed and focused on commercially viable infrastructure projects; • Environmental and Social Safeguards (ESS) system well institutionalised with proactive approach-based activities; • Global standards-based best practices applied in its whole operations – providing credit, risk management, good corporate governance, and ESS.
PT SMI (Executing Entity for Component 1)	• Local Financial Intermediary under Component 1; and • Monitoring, Supervision and Evaluation of the Final Beneficiaries' implementation under Component 1. • Goal-oriented government entity working towards accelerating national infrastructure that is well-versed in mobilising resources from varying institution types to finance infrastructure projects;

		Experienced in sustainable development and climate resilience (e.g., water, renewable energy generation, transport, and agriculture-related) projects for the Indonesian community.
	PT KEB Hana Indonesia (Executing Entity for Component 1)	- Local Financial Intermediary under Component 1; - Monitoring, Supervision and Evaluation of the Final Beneficiaries' implementation under Component 1; and - Implementing Entity (in-kind contribution) under Component 3.
		- Able to utilise and leverage lessons learnt from the most extensive global network among commercial banks in Korea across 24 countries; - Showcasing good business ethics and sound corporate governance at both the global and local levels through its strong internal controls and monitoring system.
	PT Bank IBK Indonesia (Executing Entity for Component 1)	- Local Financial Intermediary under Component 1; - Monitoring, Supervision and Evaluation of the Final Beneficiaries' implementation under Component 1; and - Implementing Entity (in-kind contribution) under Component 3.
		- Government entity with unmatched experience across Korean commercial and government financial institutions in financing SMEs, microbusinesses, and start-ups in Korea and abroad; - Leader in promoting government policy related to SMEs and ready to share knowledge and best practices through the Programme; - Established and in-house ethical management and human rights management systems meeting globally recognised ESG standards.
	KB Bukopin (Executing Entity for Component 1)	- Local Financial Intermediary under Component 1; - Monitoring, Supervision and Evaluation of the Final Beneficiaries' implementation under Component 1; and - Implementing Entity (in-kind contribution) under Component 3.
		- Largest Korean bank by assets, with substantial financing experience in SME loans through its breadth and depth of quality asset portfolios to share, and longstanding focus on the Southeast Asian market; - Ranked first in the Dow Jones Sustainability Index World for six consecutive years to indicate its commitment to creating sustainable value, healthy corporate governance, and promoting responsible management for the environment and society.
	PT Bank Shinhan Indonesia (Executing Entity for Component 1)	- Local Financial Intermediary under Component 1 - Implementing Entity (in-kind contribution) under Component 3 - Monitoring, Supervision, and Evaluation of the Financial Beneficiaries' implementation under Component 1
		- Equipped with diverse financial services and qualified employees ready to serve the public in need of access to finance - Recorded a meaningful growth even amid COVID-19, due to its strategic business management under the proper guidance of Shinhan Financial Group, Korea's one of representative financiers

	Tracks, monitors, and evaluate its corporate governance for the implementation of good corporate governance, along with a strict level of risk and compliance management practices.
PT Bank Woori Saudara Indonesia (Executing Entity for Component 1)	- Local Financial Intermediary under Component 1 - Implementing Entity (in-kind contribution) under Component 3 - Monitoring, Supervision, and Evaluation of the Financial Beneficiaries' implementation under Component 1 - Proven financial system and experience in corporate finance, along with a growing market share and net growth in Indonesia, fundamentally based on supervision of Woori Financial Group, Korea's one of leading commercial and investment bank. - Good Corporate Governance system settled down bankwide, along with the risk management and internal control system. - Has a focus on sustainable finance for contributing to the Sustainable Development Goals (SDGs), through financially supporting environmental, social, and educational areas in need.
ASEAN Centre for Energy (Executing Entity for Components 2 and 3)	- TA (Technical Assistance) Service Provider under Components 2 and 3 in partnership with Experts in various fields. - Proven capacity as an inter-governmental organisation in terms of expertise in energy, policy advisory services, and various knowledge sharing and capacity building activities across ASEAN countries; and - Equipped with relevant data and information accumulated through various policy analyses, energy assessments, and studies on best practices and lessons learnt from cross-country initiatives in pursuit of sustainable and low-carbon development in ASEAN.

Governance and Implementation Arrangements of the Programme. For the effective implementation and sustainable management, the Programme structures different governance and implementation arrangements by component. Component 1 features the private investment element whereas the Components 2 and 3 are composed of TA activities supported by the GCF grant proceeds in collaboration with various partners, such as BASE and Korea Energy Agency (KEA) equipped with decades of extensive experience and expertise in energy efficiency business areas. In particular, KEA is actively conducting international co-operation projects such as establishing Infrastructure of Climate Business, laying the foundations for energy efficiency projects and supporting policy framework in developing countries, based on over than 40 years of experience and knowledge on domestic energy projects. The following implementation structures are indicative, subject to change as per addition of additional collaborating partners whom with on-going discussions on partnership has continued.

KDB would conduct periodic meetings between KDB, each of FIs, and the Guarantee Advisor under Component 1, so as to i) exchange the market information, ii) share any feedback or concerns arisen from the energy efficiency financing activities, and iii) discuss the existing or new sub-projects from the portfolio management perspective. For clarity, these meetings are not for a decision-making for FIs' lending activities. FIs have full authority to select, appraise, structure, execute, and monitor each of their sub-projects, provided that those activities and processes are fully in line with the Eligibility Criteria and the KDB and GCF's Guidelines.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Advanced Financing Scheme to Overcome Low Demand and Financial Barriers. Local FIs shared that a lack of liquidity, is not a primary concern for Indonesian financiers considering participation in energy efficiency finance or other sectors in general. Securing financing through global banking or capital markets on competitive terms, aside from the loan funding from the GCF, is accessible without substantial barriers given the high credit worthiness of Indonesian financial institutions. Instead, a common concern of prospective FIs and consideration of the Indonesian Financial Services Authority (OJK) is that the perceived associated risk against borrowers and/or energy efficiency projects and technologies is too significant. Though a number of prospective FIs have been interested in providing loans and other financial services related to the energy efficiency market, the lack of historically successful energy efficiency projects reveals a gap between the acceptable risk level for FIs for energy efficiency projects and the current risk level of financing such projects. To address this barrier, the Programme was curated to provide a credit guarantee cover mechanism from the GCF for FIs to mitigate the risk gap that they would face as they invest into the evolving energy efficiency market, rather than directly providing loan capital. With GCF providing guarantees cover for FI own loans that meet the necessary criteria outlined further as mentioned in B.3, FIs would now be able to accept any credit risk level of final beneficiaries, in particular those of the least creditworthy SMEs so that those beneficiaries can access energy efficiency loans and begin to expand the market.

The GCF Credit Guarantees Cover Agreement will also diversify the existing energy efficiency financing efforts, as well as GCF's portfolio of financing, as energy efficiency loans are the most widely discussed energy efficiency financing model, with financial institutions, primarily banks, issuing loans to final beneficiaries for projects (DeT'serclaes, 2010; Kats et al., 2011; MacLean & Purcell, 2014; Würtenberger, 2012). Climate finance research suggests that financing for energy efficiency by private actors only accounts for 4% of the total investment into climate finance of USD 13 billion between 2015 and 2018, reflecting the deep need and significant opportunity to pursue an increase in energy efficiency investment in Indonesia. Because direct bank loans to existing customers with stringent lending terms, rather than potentially new borrowers with higher credit risk levels, are customary of the loan market (Würtenberger 2012), GCF's role in providing support to bridge the risk gap for FIs and ultimately providing more competitive financing terms to final beneficiaries for energy efficiency projects is key.

Based on the information above, KDB developed the concessional GCF Credit Guarantee Cover portion of the Programme as its primary financial component. As detailed in B.3 and B.4, the Programme does not simply provide low-interest funding to final beneficiaries but provides credit risk guarantees for FIs for their own loans, providing a higher guarantee coverage ratio to those with lower credit, which leads to increased energy efficiency financing and addressing of the market in Indonesia.

Justification for the Concessionality Level. Commercial interest rates of a typical corporate loan in the local financial market range between low-teen to high-teen percentages, although the rates vary greatly depending on the creditworthiness of the borrower, nature of underlying investments and risk allocation structure, detailed financing terms including a tenor, and/or the current market situation. The Market Assessment of the Programme shows that the rates often increase by more than 15% for financing the deployment of energy efficiency technologies compared to standard rates.¹⁷ The increased rates are not attractive for industry enterprises and technology providers to utilise their existing and limited credit lines available for an investment that may not directly support their core business areas nor immediately increase output production and revenues. Thus, commercial loans in Indonesia are currently not a feasible option for

¹⁷ Bank of Indonesia. 2020. Interest rate of rupiah loans by group of banks and type of loans. Available at https://www.bi.go.id/seki/tabel/TABEL1_26.pdf. Accessed on 23 April 2020.

industry stakeholders to secure the required private ‘buy in’ to take off in the local energy efficiency market. In addition, market players – particularly purely domestic firms and/or institutions – are relatively small with weak financial positions. As a higher risk premium on lending is imposed on vulnerable borrowers, it is challenging for the vulnerable business group, mainly local SMEs, to access and secure the finance required for manufacturing and installing energy efficiency technologies.

Considering Indonesia’s low electricity tariff and industry’s perception and knowledge of the energy efficiency business, analysis and stakeholder feedback indicates that a low interest rate of approximately less than 10% is needed to induce financial institutions and industries to participate in energy efficiency projects.

As such, final interest rates to the local end beneficiary group are assumed to be a single-digit percentage with a maximum up to a low-teen percentage, after taking the GCF Credit Guarantee Cover, which represents a margin from FIs and other financing costs, into consideration. This scenario is perceived as competitive compared to the market terms available in Indonesia.

Most importantly, the Programme ensures that the concessionality and guarantee coverage ratio, of up to 95% depending on the size of final beneficiaries, offered by the GCF is passed on to the end beneficiaries, such as SMEs via local FIs, imprinting the lessons learnt; concessionality often disappears as international funding comes through multi-layered intermediaries to final beneficiaries. The Programme will be equipped with protection and monitoring mechanisms so that margin and/or financing costs from the FIs, which are included in the ultimate rates applicable to end beneficiaries, will be capped in contract under KDB supervision. In addition, with the successful implementation of financing based on the concessionality provided by GCF, confidence over energy efficient technologies will be accumulated amongst industries and financiers, and financiers will have a better understanding on energy efficiency financing and a credit analysis and mitigation system internally, which will make financial institutions have a willingness towards energy efficiency financing even after the Programme is over. Finally, if these successful cases continue, we can expect other private investors currently under the surface will join energy efficiency financing.

Rationale for the Proposed Solution Financing Package - TA Grant Facility. The Programme never disregards how the current risk averse trajectory of the energy efficiency market has been long formulated by a lack of awareness, knowledge, capacity as well as misperception of relevant stakeholders, and why it is still getting stuck despite some trials. Even if a solution financing scheme is prepared, it would be too hasty to expect the market to be boosted by prompt actions from both private and public stakeholders. Relevant interventions along with the pioneering energy efficiency funding is therefore inevitable. As such, technical assistance activities are planned with the support of the GCF grant instrument, as well as in-kind contributions from other participants including KDB.

In this context, de-risking mechanisms can play a critical role in assisting market functioning for investments that are considered too and insubstantial to be attractive for banks to take on or too challenging to analyse the related credit risks, such as energy efficiency markets and the SME market (International Finance Corporation 2013). The Programme suggests two kinds of de-risking mechanisms in Component 1 and Component 2 as Table 11 shows below. These two schemes can work in tandem with each other and create synergies to increase energy efficiency in the Indonesian industrial sector.

【Table 11】 De-risking Mechanisms in the Programme

Component 1 (financing component)	Credit Guarantee for Sub-project loans between LFIs and final beneficiaries
Component 2 (TA component)	Introduction and development of ESI (Energy Saving Insurance) scheme

	Development of customised ESCO finance structure with guarantee scheme
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Limited Public Endeavours and the COVID-19 Impact. The government of Indonesia and the public sectors have carried out limited studies and have developed regulatory frameworks for promoting energy efficiency in the industry sector. Such endeavours, however, have not aggressively pushed the deployment of the industrial energy efficiency technologies or the implementation of the relevant businesses, and anticipated benefits of energy efficiency have not been translated into energy intensive industries.¹⁸

Furthermore, as the economy has contracted and struggled amid the prolonged pandemic crisis, the national indebtedness has increased due to the government's borrowing of funds from international funders for the purpose of disseminating COVID-19 relief packages and economic stimulus towards the post-pandemic recovery, according to Bank of Indonesia. Confronting tighter budgets, particularly in the midst of the unprecedented COVID-19 crisis, the government and the public sector are more likely to have a priority of funds allocation towards the post-pandemic recovery schemes over energy efficiency initiatives.

The Importance of the GCF as an Anchor Investor. Surrounded by myriads of barriers, constraints, and unfavourable local circumstances, the GCF ought to play a crucial role as an anchor investor with a financing solution package. In this regard, the GCF is a sole solution financier, particularly considering its strategic thematic priority to 'industries and appliances' for driving a low-emission development pathway in emerging economies, and with financial flexibility and mobilisation capacity.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Function as a Catalyst to Drive a Genuine Paradigm Shift towards a Sustainable Energy Efficiency Financing and Markets. The main financial barriers are assessed by Market Assessment as i) a lack of understanding/capabilities of local FIs, and ii) a lack of credit risk mitigation techniques/mechanisms. Through the successful implementation of energy efficiency financing under component 1 and TA activities including the development of de-risk mechanisms under Component 2, LFIs can have a better understanding on energy efficiency financing and a better de-risking mechanisms within their internal system, which will make LFIs keep joining energy efficiency financing even after the Programme is over. In other words, the Programme will function as a catalyst to drive a genuine shift through enabling environment formation, beyond one-off financing provision.

In addition to financing and de-risking scheme development, TA services will be provided to various stakeholders including industry players, financial institutions, and government authorities. With the successful implementation of TA activities under component 3, local stakeholders can accumulate their confidence over energy efficient technologies and financing, which means that a sustainable energy efficiency market including locally fit de-risking scheme and regulatory framework development will be created in Indonesia industry sector.

Formulation of the Market and Enabling Conditions as the Key Post-Programme Sustainability Strategy. The Programme ensures its post-Programme sustainability after the exit of the GCF and KDB, by making its best efforts to formulate the energy efficiency market surrounded by enabling circumstances. In short, its exit strategy implies the relevant market creation with enabling settings, not to make it passively left dependent on the external support – Indonesia is an attractive emerging market. Thus, international cooperation trials for energy efficiency are most likely to continue – but to bring lasting actions amongst local players. Once the market is formed, a working mechanism where ambitious pioneers emerge to the surface

¹⁸ Asia-Pacific Economic Cooperation (APEC) Secretariat. 2017. Energy Efficiency Finance in Indonesia: Current State, Barriers and Potential Next Steps - APEC Energy Working Group

and somebody is willing to bear the roles and responsibilities that will allow the market to settle down. The Programme sees the exit strategy as the on-going process of deploying this market-driven sustainability principle in following through all stages of the Programme cycle with often making a necessary modification to the plan if needed over a decade.

【Table 12】 Four Channels of Interventions for Market Creation

Market Enabling Create the frameworks within which markets can function, including policy, regulation, platforms, financial, and other interventions. E.g. The Programme will make sustainable energy efficiency framework from the policy, financial, and technical perspectives.	Innovation and Competition Promote competition through innovation, improved management, cost reductions, and efficiencies that encourage other market players to continue improving their products and services .
Demonstration and Replication Provide demonstration effects, encouraging replication and more generally the spillover of ideas. .	Capacity Building Build capacity and skills and create standards that open new market opportunities and support market development.

(Source: IFC website)

Taking difficulties of market creation, particularly in the emerging economy context, into account, the International Finance Corporation (IFC) usually takes multiple interventions that contain upstream activities and market enabling work at the sector level. For the sake of the market development, the Programme also takes coordinated interventions that integrate the private investment financing scheme and TA activities such as capacity building of key local stakeholders and knowledge sharing and replication across the ASEAN region.

Harnessing the Programme as a Golden Opportunity to Bring Institutional Changes of Indonesian Partners for Sustainability. Undoubtedly, ‘sustainability’ is an important value, so it is a must agenda throughout global discourses on international cooperation projects in the developing world. It is therefore familiar with many, but never easy to materialise on the ground as a matter of fact. The Programme, as an ambitious trial supported by the largest multilateral climate fund, shall take the sustainability issue into account throughout the industrial energy efficiency lifecycle, and even after the exit of the GCF and KDB.

Above all things, as an accredited entity with accountability on the Programme success, KDB shall try to create an enabling environment in which not only the Programme but also other relevant businesses can self-sustain without external capital injection. For realising this, formulating and facilitating the market is the kernel of the energy efficiency initiative. Before the launch of the Programme, at present, the Indonesian market possesses a few potential players and they have been barely active under the surface while confronting the existing stumbling blocks and obstacles. After the multi-dimensional intervention of the Programme, Indonesia will see the market that works along with enhanced institutional capacity. In short, technical assistance for market readiness (particularly under Component 3) is an important factor that will function as an exit strategy for the sake of the post-Programme sustainability, eliminating dependency on concessional financiers and donors. Besides, the Programme tries to implant other supplementary tools and measures from its beginning to the end, delivering a complete functional solution to ensure the industrial energy efficiency in a sustainable manner.

Strategically Crafting Climate-responsive Recovery in the Post COVID-19 Period for Sustainability.

As tried elsewhere in most other countries, Indonesia has been rolling out its stimulus packages in response to the pandemic outbreak and economic recession. The Programme strategically crafts the way of integration into the national stimulus package so that the Programme could self-sustain as a national recovery measure after the exit of the GCF and KDB, not as a one-off event. Of the economic measures, the Indonesian government has devised a stimulus for SMEs like interest subsidies for microcredit, MSME financing, guarantees, and placement of funds in banks and the Programme will make endeavours to come up with ideas for the GCF-funded activities to be integrated into regular government exercises. The way of integration – in what form and what specific measures – is to be brainstormed and discussed with relevant government stakeholders for a genuine country ownership. Such integration of the international support Programme into national planning is an effective exit strategy for the Programme to self-sustain after the implementation period ends.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	105			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(iii) Equity	<u>Enter amount</u>			<u>Enter % equity return</u>		
(iv) Guarantees	100	10 years				
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	5					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	142.7			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Local FIs ¹⁹	<u>Senior Loans</u>	<u>133.33</u>	<u>million USD (\$)</u>	<u>10 years</u> <u>3 years</u>	<u>Enter%</u>	<u>pari passu</u>
	<u>In-kind</u>	<u>0.29</u>	<u>million USD (\$)</u>	-	-	-
Local SMEs	<u>Equity</u>	<u>9</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
KDB (AE)	<u>In kind</u>	<u>0.03</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Korea Energy Agency (KEA)	<u>In kind</u>	<u>0.04</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Korean Banks	<u>In kind</u>	<u>0.01</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Total financing (c) = (a)+(b)	Amount			Currency		
	<u>247.7</u>			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	N/A					

¹⁹ Participation amount by each FI is undecided

C.2. Financing by component							
Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
<u>Component 1</u> Energy Efficiency Finance	Increased volume of energy efficiency financing in Indonesia industry with an increased participation of the local FIs in energy efficiency financing	<u>242.33</u>	<u>100</u>	<u>Guarantees</u>	<u>133.33</u>	<u>Senior loan</u>	<u>Local FIs</u>
					<u>9</u>	<u>Equity</u>	<u>Local borrowers</u>
	-M&E	<u>0.29</u>			<u>0.29</u>	<u>In-kind</u>	<u>Local FIs</u>
<u>Component 2</u> De-risking Mechanisms	Generation of Locally customised ESI Mechanisms	<u>1.21</u>	<u>1.21</u>	<u>Grants</u>			
	Developed ESCO financing stricture	<u>0.36</u>	<u>0.32</u>	<u>Grants</u>	<u>0.04</u>	<u>In-kind</u>	<u>KEA</u>
	M&E	<u>0.06</u>	<u>0.06</u>	<u>Grants</u>			
<u>Component 3</u> Technical Assistance (TA) for the Market Readiness	Enhanced capacity of financial institutions	<u>0.97</u>	<u>0.93</u>	<u>Grants</u>	<u>0.03</u>	<u>In-kind</u>	<u>KDB</u>
					<u>0.01</u>	<u>In-kind</u>	<u>Localised Korean Banks' subsidiaries</u>
	Generation of market preparedness of industry stakeholders	<u>0.89</u>	<u>0.89</u>	<u>Grants</u>			
	Polished and strengthened regulatory framework policy that incentivises energy efficiency stakeholders	<u>0.75</u>	<u>0.75</u>	<u>Grants</u>			
	Spreading positive perceptions	<u>0.69</u>	<u>0.69</u>	<u>Grants</u>			

	of energy efficiency finance in ASEAN region						
	M&E	0.15	0.15	Grants			
Indicative total cost (USD)		247.7	105		142.7		

Main Terms and Conditions under Component 1 (See Annex 14 for details)

1. Tenor: The maximum tenor will be the shorter of ten (10) years and the period remaining between the signing of the AE Guarantee and the Completion Date
2. Currency: USD
3. Purpose of Guarantee : To guarantee FI Own Loans that meet the Eligibility Criteria
4. Guaranteed Loan : Each FI Own Loan shall rank pari-passu with other senior loans of the Borrowers
5. Guaranteed Risk : Any default of the Scheduled Payment by the relevant Final Beneficiary under the Sub-Project Agreement which remains unpaid 25 business days after becoming due.
6. Scheduled Payment

Each (a) repayment of principal of the FI Own Loan, and (b) payment of Covered Interest that is due and payable by the Borrower (i) on the Scheduled Payment due dates and in the amounts specified in the Payment Schedule for the FI Own Loan, or (ii) by acceleration or mandatory prepayment caused by default on the scheduled principal and interest payment.

The Scheduled Payment excludes any break or unwinding costs upon any prepayment of the FI Own Loan, any default interest, penalty, fees, costs, or any gross-up payments which are not expressly covered herein.
7. Guarantee Claim (Payment Demand)
 - i) Not later than [30] days after the occurrence of the Guaranteed Risk, the Beneficiary may make demand on the Guarantor for the Guarantee Payment ("Demand Date") by written notice in substantially a pre-agreed form ("Demand Notice"). The FI shall confirm that it has not recovered the defaulted payment since the Demand Date.
 - ii) The Guarantor shall pay the relevant amounts in USD as specified in the Demand Notice. For the avoidance of doubt, the payment from the Guarantor to the Beneficiary is limited to the amount that GCF has paid to the Guarantor under the relevant Guarantee Cover Arrangement.
 - iii) Upon any payment made by the Guarantor in respect of any claims under the Guarantee Agreement, the Guarantor shall be subrogated to all the rights and remedies of the Beneficiary against the relevant Borrower in respect and to the extent of the sums which the Guarantor has paid pursuant to the Guarantee Agreement.
8. Obligation of each FI
 - i) Each FI must take all necessary measures in good faith with the same level of care exercised towards other assets not covered by the Guarantee Agreement, and notify the results of such measures to the Guarantor.
 - ii) If each FI becomes aware of any circumstance of any default, or any event that may damage the ability of Sub-project or any person providing a security mechanism from fulfilling its debt obligation, or that an event of default has occurred under a Sub-Project Loan agreement, each FI must notify the Guarantor in writing within one (1) month after the date when each FI became aware of such event.
 - iii) Each FI must take all necessary measures to collect the principal and interests under the Sub-project Loan Agreement, compensation for losses, penalties, and other similar amounts, even after each FI has received the Guarantee Payment, and notify the result of such measures to the Guarantor.
 - iv) Any collected amount calculated as the amount collected through iii) above multiplied by the Guarantee Coverage Ratio under each Guarantee Agreement shall be paid to the Guarantor only up to the amount of the Guarantee Payment received, within fifteen days after the date when collected.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

Improved Capacity to be Built for Implementation. The Programme finances capacity building activities to support a successful implementation of the Programme. Despite quite many capacity building activities for industries, ESCOs, and/or government, the past activities had confronted implementation challenges with still an under-served demand in institutional, financial, technological, and regulatory capacity enhancement. Indonesia needs to keep out of the past trajectory of short-term trainings and capacity building programmes on a stand-alone basis. In the absence of an enabling circumstance, the Programme will repeat the past failures; it cannot materialise its activities nor achieve the impacts.

Lessons learnt from the precedents are carefully taken into account when it comes to designing TA activities; (lesson 1) capacity building towards industries should eye on the top management level since knowledge sharing to working level technicians and managers rarely translate into practice and a strategic determination from the higher ups who tend to prefer a short-term hit on profitability; (lesson 2) capacity building towards financial institutions should be carried out with selected local FIs on a learning by doing basis over the course of actual pipeline development, financing, portfolio management, and repayment under Component 1; and (lesson 3) the regulatory support for the government, ministries, and public agencies should never disregard the inter-sectoral policy synchronisation and collaboration amongst themselves, in that each of them has individual insights and existing institutional systems.

In this regard, the Programme sets out capacity building activities under Components 2 and 3, as seen in B.3, with the support of GCF grant proceeds and in-kind contributions from an executing entity, alongside the market-driven financing under Component 1. The two TA components seek to enhance the market, institutional, and regulatory capacity of Indonesia where the industrial energy efficiency financing schemes are to be deployed. It further expands the regional capacity beyond Indonesia through regional knowledge sharing and dissemination activities of the energy efficiency financing schemes, given the connected financial network across ASEAN.

Financing Technology and Innovation. From a technological aspect, the Programme will accelerate the development and adoption of new energy efficient technologies in the local industrial sectors. This embraces a great potential of formulating a new market segment that promotes relevant technology research and development, production, installation, operation and maintenance service, and monitoring and evaluation expertise. Additionally, expected engineering and/or technological modifications to the existing industrial processing procedures are a new practice – i.e., innovation – to Indonesia. The Programme could also bring about a revival of ESCO business via the proposed de-risking mechanism under Component 2.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Greenhouse Gas (GHG) Mitigation Impact Potential Based on a Programme of Activities (PoA) Approach. The Programme has a direct GHG mitigation impact potential estimated at approximately 3.1 million tCO₂eq for the lifespan of the technology. This results from the adoption of the PoA approach that clings to principles of the Clean Development Mechanism (CDM) methodology in terms of baseline comparison and project emissions by following eight steps below. See Annex 22 for details.

[Table 13] PoA-based Calculation on GHG Emission Reduction

[Step 1] Deriving Emission Factors

- Different fuel sources are used for individual technologies; boilers use gas: and motors, VSD, pumps, fans, and WHR use electricity. 56.1 (eq kgCO₂/GJ) is an emission factor for natural gas as per GHG Emission Factors of Electric Power Interconnection System 2016 by the Ministry of Energy and Mineral Resources in Indonesia (IPCC). As for electricity, a grid emission factor is 0.675 (UNFCCC, IFI Dataset, Version 3.2).

[Step 2] Confirming Industrial Energy Use and Energy Consumption by Technology

- Energy use in two major industrial energy systems – motor and steam systems – was estimated for Indonesia, in addition to the above analysis.

[Step 3] Confirming Technologies and the Number of Units within each Bundle

- Bundle A – focus on motors with VSDs, pumps, fans, and boilers, such as in food and beverage, and manufacturing sectors
- Bundle B – focus on similar technologies but also adding a chiller as a cooling technology for petrochemical sector, for instance
- Bundle C – Cases of loans taken by a technology provider who would serve SMEs

[Step 4] Confirming Annual Emission Reduction of Each Technology within a Bundle

- The annual GHG reduction (eq tonCO₂/yr) was calculated through annual energy savings per technology multiplied by the number of units per technology within each bundle – A, B, C and WHR
- * It is assumed that actual energy savings is 55% of the one from market assessment

[Step 5] Combining Emission Reduction Estimates of All Technologies

- The annual GHG emission reduction of each bundle is calculated by a sum of all annual mitigation figures of each technology within a bundle. The value was multiplied by the grid emission factor of Indonesia afterwards.

[Step 6] Calculating Emissions Reduction of Bundled Projects as per Disbursement Schedule

- Disbursement Schedule (Availability period of 3 years)

Type of funding	Disbursement schedule										TOTAL (million USD)
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	
Loan (Covered)	33.3	33.3	33.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	100.0
Loan (Uncovered)	11.1	11.1	11.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	33.3
Equity	3.1	3.1	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.4
Yearly total disbursement	47.5	47.5	47.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	142.7

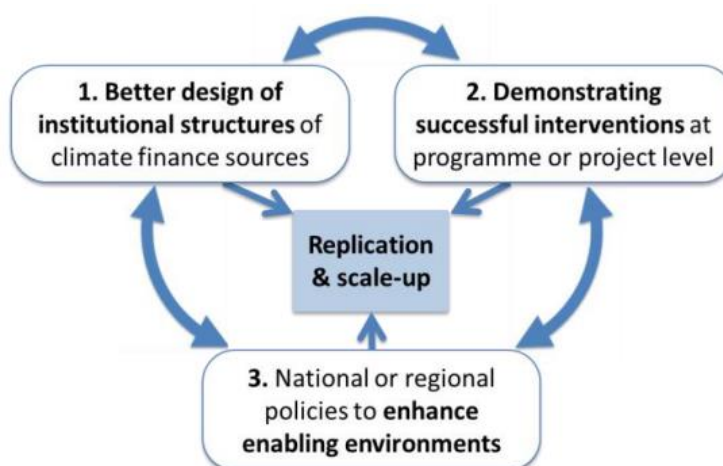
[Step 7] Concluding the Impact Potential of the Lifespan in Total

- A sum of emission savings of all bundles is concluded as the GHG emission reduction with the machine lifespan, which is assumed as 10 years for all technologies from conservative perspective.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Scaling Up and Replicating the Effective Energy Efficiency Interventions. The Programme recognises the importance of scaling up and replicating energy efficiency business trials so as not to be a one-off initiative. The OECD identifies three key aspects for scale up and replication of climate finance interventions that were carefully considered during the designing phase: (1) a better design of the institutional structure and decision-making framework; (2) demonstration of successful interventions; and (3) policy support to enhance enabling environments.²⁰ The Programme has components that reflect on the three aspects so that it can bring energy efficiency activities in both sufficient quantity and quality in Indonesia.

【Figure 17】 Three Key Aspects for Scaling-up and Replication of Climate Finance Interventions (OECD)



As for the first aspect of designing a better institutional structure, the Programme takes a two-step financing structure that simultaneously utilises KDB's capacity, as a global financier with abundant energy efficiency finance experiences and local FIs with a strong local presence and expertise in allocating and disbursing the funds, under the government (including NDA) as a regulatory standard setter. With regards to a success demonstration, the Programme aims to be the first exemplary best practice by additionally devising energy efficiency guarantee financing beyond the conventional concessional loan that has been touched by preceding trials. Such a diverse flexibility of finance will definitely boost the success potential of the Indonesian-led energy efficiency move. Concerning the policies to enhance enabling environments, the Programme embraces the fine-tuning of the local regulatory setting as one of critical TA activities.

Designed as a Catalyst for Formulating Enabling Environments, Driving a Genuine Paradigm Shift.

The Programme dedicates its components to improve awareness and dissolve barriers identified through the Market Assessment and a series of consultations; it equips itself with systemic TA supports on top of a huge financial support. In short, the Programme reminds of the past precedents that financing solutions – designed by global players – were not easily adopted nor implemented on the ground. Therefore, diverse activities such as capacity building, knowledge sharing, and policy recommendations for the government are included in the TA components so that the primary component's (Component 1) financial incentives for local borrowers genuinely would work in the market.

The Programme plans to function as a catalyst that brings the eyes of the market into the brand-new energy efficiency financing schemes so as to be more sophisticated and advanced to next steps. The Indonesian

²⁰ OECD. 2014. Scaling up and replicating effective climate finance interventions.

financial market is already active and commercially oriented. Once energy savings insurance picks up on the notion of a new product, the market will most likely jump in to provide similar products. Meanwhile, the market and banks will be more familiarised with investment in energy efficiency. This will lead to lower interest rate of loans enticing increased demand for these financial mechanisms. The Programme further anticipates that the whole ASEAN region would have an eye on this, developing their own de-risking mechanisms customised into their respective market circumstances; this is a genuine regional paradigm shift towards an energy efficient development pathway.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Green Jobs Created and Sustained with the Energy Efficiency-driven Economic Leap. It expects to bring about economic co-benefits. The local job creation – especially green jobs – with income generation is the major economic benefit to the country, in addition to energy savings-caused profits in industries. The energy efficiency business plans the employment of professional and administrative human resources to install, manage, monitor, evaluate, and maintain the energy efficient appliance and equipment in factories and plants. It assumes considerable economic impacts given a pool of several hundreds of energy managers and 30 ESCOs currently registered across Indonesia. The emergence of green jobs created and sustained will relieve unemployment as much as scale up and replication of the Programme proceeds.

Transition towards an energy efficient economic structure will result in great economic productivity and long-term economic prosperity, making the country more competitive in the ASEAN economy. A recent study reveals that firms whose energy productivity – economic output per unit of energy use – is the worst in industry could further achieve growth in annual profits at a rate of 2.2% to 13.8% by bringing energy efficiency into the industrial processing line.²¹ Assuming such room for potential growth, the Programme is integral to endeavours to jumpstart Indonesia.

Gender Co-benefits around Skilled Female Workforce and Women-led SMEs. Very well aware of tremendous merits of women-empowered advancement in line with the energy efficient economy booming, the Programme submits gender assessment and action plan in both Programme and project levels. Since the inception stage, KDB gender and social development specialist therefore has engaged the concept development in pursuit of a holistic gender mainstreaming. Taking into account the two-step financing structure of the Programme, as the Accredited Entity, KDB emphasises the incorporation of a women-led SME framework and women's empowerment through the technical and vocational training provision that seeks for higher skilled occupation of women on the energy efficiency labour market. Detailed activities are specified in G.2 (Gender assessment and action plan section). Gender co-benefits further relate to social co-benefits. Proactive women's empowerment activities will turn out to produce positive externalities in diverse social areas, such as healthcare and education, referring to internationally proven cases that the skilled female workforce will gain confidence and voice in the household decision making process and their generated income could be invested in schooling of their children and nutritious food of their families. To put it succinctly, the positive externalities are to be extended beyond gender and relevant arenas.

Achieving SDGs with the Far-reaching Energy Efficiency Intervention. Energy efficiency is one of the best buys in global energy and has a crucial role in achieving the SDGs. As prescribed above, the Programme expects positive externalities in various aspects, implying its direct and indirect contributions to all of the 17 UN SDGs. Nevertheless, focusing only on a direct linkage, the Programme will support Indonesia mainly to contribute to the global targets: Goal 5 (gender equality), Goal 8 (decent work and economic growth), Goal 9 (industry, innovation and infrastructure), Goal 12 (responsible consumption and production), Goal 13 (climate action), and Goal 17 (partnerships for the goals). As a far-reaching energy intervention, it definitely reflects the critical ethos of the SDGs.

²¹ ClimateWorks Australia, 2016

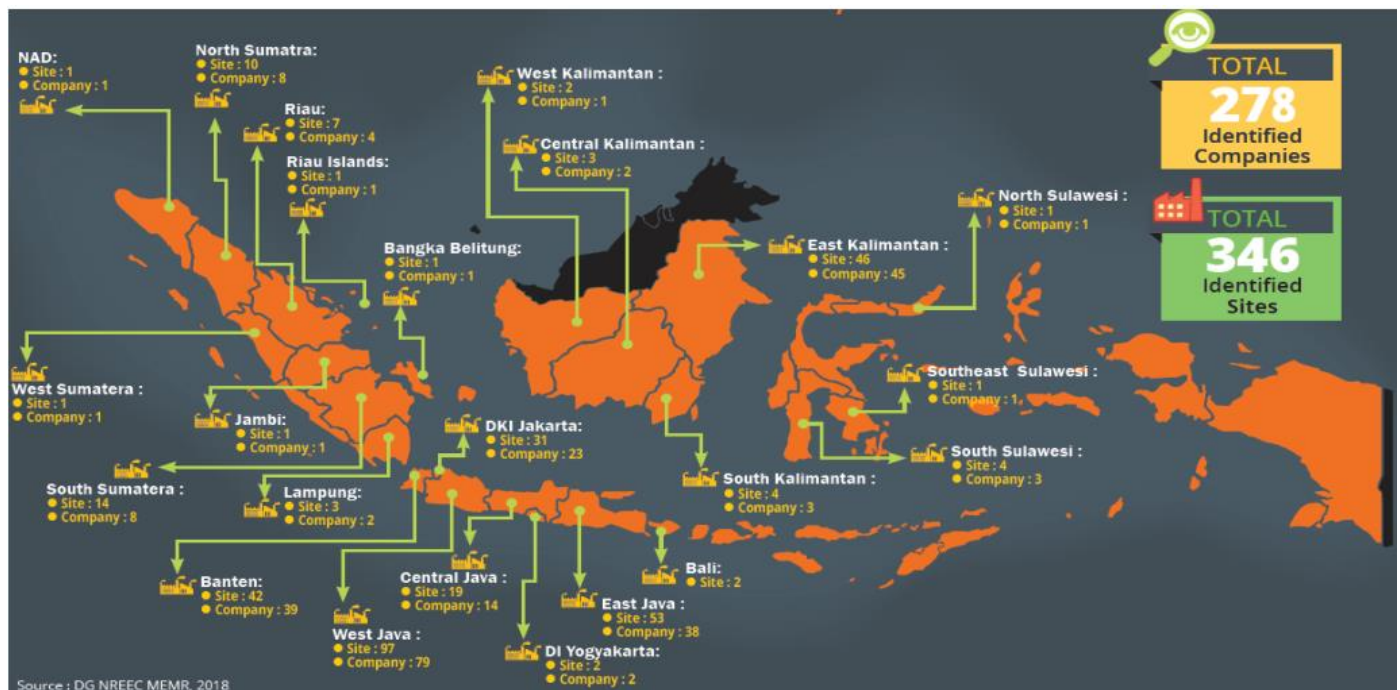
[Figure 18] Direct Contribution to the 6 SDGs



D.4. Needs of recipient (max. 500 words, approximately 1 page)

Need-based Targeting: Mapping the Needs of Energy Efficient Technologies across Indonesia. Confronted against a surging energy demand amid the economic growth and urbanisation, the deployment of energy efficient technologies is a desirable pathway ahead of Indonesia to meet the surging needs and ensure energy security in coming decades. As a result of mapping the needs of industrial energy efficiency enhancement, the Indonesian MEMR has identified potential target locations with energy intensive companies with more than 6,000 TOE of energy consumption: 278 companies and 346 sites, mainly spread out throughout four major provinces – West Java, East Java, and East Kalimantan, and Banten. These four provinces account for 70% of the total industries obliged to do energy management such as appointing an energy manager and implementing energy programmes with audit activities, as regulated by the Government Regulation on Energy Conservation no. 70/2009. Due to aforementioned obstacles, it has barely materialised in Indonesian industries, in actual. With reference to the MEMR mapping as of 2018 (see Figure 19 below), the Programme will help Indonesia take actions and prioritise relevant policy and regulatory setting, so as to track a genuine progress towards the country's policy goals and international climate commitments.

[Figure 19] Mapping Prospective Target Beneficiaries



(Source: Indonesia MEMR)

Needs to build an Energy Efficiency Financial Ecosystem enabling Energy Efficiency Market even after the Programme. The Indonesian Industry has a need to establish a sustainable financial ecosystem

between private financial institutions and local players, even when there is no participation of international financiers such as MDBs and international organisations (IOs) with grants, concessional loans, or risk-taking mechanisms. For the energy efficiency market to be active even after the Programme ends, the Programme was designed to include 1) project replication and scale up trials through successful pilot programme demonstration and 2) development of customised de-risking mechanisms for Indonesia leading more attractive risk-return trade-off in energy efficiency finance.

Urgent Need for Strengthening Institutions and Implementation Capacity, in the Midst of Acute and Chronic Climate Impacts. As mentioned throughout the previous sections, Indonesia is undoubtedly climate vulnerable and its needs for energy efficient measures are crystal clear. An extensive use of inefficient technology, equipment, and processes in multiple industrial sectors put Indonesia in a more vulnerable position. The potential direct and indirect mitigation impacts of this Programme fit well with the country needs.

Final GHG mitigation impacts aside, the Programme also has eyes on insufficient institutional and implementation capacity of Indonesia. The local energy sector lacks technical capacity in implementing innovative and system-based technologies. The Programme's deployment of highly efficient technologies with energy management systems and relevant regulatory frameworks, such as rating and audit systems will benefit the country, transferring technology and expertise and increasing the nation's overall capacity in terms of innovation in energy. As described hereunder, the Programme is designed based on the lessons learnt from the precedents, in line with a strong willingness of the Indonesian government to strengthen and refine the regulatory framework and policies for promoting industrial energy efficiency in consideration with synergy impacts alongside diverse relevant industrial development initiatives.

D.5. Country ownership (max. 500 words, approximately 1 page)

Indonesia-led Initiative with a Strong Country Ownership backed by the NDA. The Programme conforms to the country ownership principle of the GCF in that the birth of the business concept was first proposed and initiated by the Indonesian NDA versed in the GCF Country Programme, and in that it is fully aligned with the national strategies, policies, and other existing frameworks of the partner country. Throughout the designing of the Programme, the Indonesian NDA, policy makers (e.g., NDA, MEMR, Ministry of Industry), private experts, and civil society organisations have proactively engaged in a series of stakeholder consultations; as developed with their recommendations from the scratch, the Programme ensures the effective implementation under guidance of the government with a strong ownership.

Alignment with the National Strategy and Regulatory Direction of Indonesia. Since 1982, Indonesia has formulated several policies and regulations to promote energy efficiency and conservation in the industrial sector. According to the MEMR Regulation No.14/2012 on Energy Management that targets the energy intensive industry, the government has imposed energy users with consumption equal to 6,000 TOE per year to assign an energy manager, plan an energy conservation Programme, perform an energy audit, and report the Programme implementation. In addition, the Government Regulation No.70/2009 states incentives, such as tax exemptions, fiscal incentives for import, a low interest rate for investment in energy efficiency. Also, OJK promotes sustainable finance in its roadmap, setting up the OJK Regulation No.51/POJK.03/2017 that requires all the major banks in Indonesia to provide a sustainability report by the end of 2019. This newly pushes the local private sector to invest in green business areas including energy efficiency. Indonesia accordingly has RUEN that stipulates energy efficiency finance to boost investments in energy efficiency with financial incentives. As explained above, the Indonesian government is clearly aiming to decrease the energy consumption while increasing infrastructure supply. However, as its scope and degree of such efforts are not sufficient to revitalize energy efficiency in Indonesia, energy efficiency market stills remains at a very initial stage, as the market assessment (annex 2) has shown,

Aligned with a willingness of the Indonesian government, the Programme recognises the government's commitment to the promotion of energy efficiency and plans to move forward in line with the national regulatory direction. The Programme therefore shall contribute to the Indonesian government's target to reduce energy intensity by 1% per year and elasticity of energy to less than 1% by 2025.

Alignment with the Nationally Determined Contribution. Indonesia submitted its NDC to the UNFCCC with a strong commitment of unconditionally reducing 29% of GHG emissions and up to 41% with international support against the BAU scenario by 2030.²² The NDC outlines the government's willingness to transform towards the low-carbon and climate resilient pathway through actions including improved energy efficiency and consumption patterns.

The Programme recognises that energy efficiency in final energy consumption is included amongst key mitigation actions in the NDC. Assuming inefficiency of the BAU scenario as a baseline, 75% and 100% of mitigation are anticipated in the unconditional scenario and conditional scenario, respectively.

Proven Capacity of KDB to Deliver Energy Efficiency Impacts in partnership with Local Executing Entities. The Programme shall benefit from the roles and responsibilities of KDB, the Accredited Entity of the Programme. When energy inefficiency was an impediment to growth so fostering its efficiency was not a choice anymore for Korea due to urbanisation, rapidly growing energy demand, and fossil fuel-based energy mix structure, KDB was at the centre of deploying energy efficiency finance nationwide. Creating and transforming energy efficiency markets on scale require tremendous efforts in addressing barriers; KDB has experience in supporting Korea leapfrog in the energy efficiency industry by tackling challenging circumstances of the then least developed market. Leveraging capacity accumulated, KDB will perform as an ideal accredited entity to support Indonesia becoming more energy efficient in collaboration with executing entities having capacity and local presence.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

The Programme aims to maximise environmental benefits, including GHG emission reduction, as well as to optimise investment returns and economic gains for local stakeholders through innovative, reasonable, and efficient financial structure and eligibility criteria.

Maximising Climate Change Mitigation Impacts. Overall, USD 100 million GCF funding will deliver mitigation benefits of USD 31.9 per tCO₂eq (assuming total machine lifetime emission reduction). Taking into account the total project cost including co-financing, the Programme will deliver climate change mitigation benefits of an estimated USD 45.5 per tCO₂eq, based on the PoA approach described in section E.2.

Leverage and Co-financing. The Programme will use USD 100 million from GCF Guarantee Cover to leverage USD 143.3 million in total project costs, including USD 133.3 million from local FIs and USD 9 millions from SMEs. The ratio of Co-financing to GCF contribution is 1.0(GCF) : 1.43(Co-financing) – this ratio is meaningful as the Programme would de-risk projects and catalyse private investment from private financiers and local industries who are yet hesitant to invest in energy efficiency areas as confronting barriers and challenges. KDB would design the following finance structure to overcome these barriers and encourage private sector participation.

Efficient Terms and Conditions. Above all, private financiers are extremely hesitant to use their own credit portfolios to fund energy efficiency businesses, with steady and acceptable rates of return on such

²² Republic of Indonesia. 2021. Updated Nationally Determined Contribution Republic of Indonesia. (submitted to UNFCCC on 21 July 2021)

investment not being assured. The kernel of the Programme is to provide incentives that attract relevant players to crowd in industrial energy efficiency trials. At present, on-going energy efficiency deals often go high-teen percentages or more than 20%. Under the Programme, a target interest rate of covered loans is expected to around 10%. Given this, favourable terms and conditions of finance are designed in pursuit of better economic benefits and bankability so that it will be able to give positive signals to other entities and industries by presenting a successful pilot project in the energy efficiency market that has not been realised so far due to lack of incentives interlinked to uncertainty and lack of awareness in the least developed market. As a result of bilateral discussions on financial terms and conditions between the GCF and KDB, the amount of proceeds and its financing terms for guarantee and covered loan has been determined at a minimum level of concessionality.

Guarantee Facility. On the ground of favourable finance provision from international financiers, the Programme shall also make an effort to catalyse and optimise co-financing opportunities via local financial institutions. The Programme therefore sets up a guarantee facility to cover FI own loans in order to facilitate commercial financial institution's participation in energy efficiency businesses. The coverage ratio of the guarantee instrument is dependent on the company size of borrowers, which will make commercial financial institutions bear some amount of credit and can be referenced in Table 6. In addition, the smaller the size of the borrower, the greater the guarantee percentage, maxing out at 95%, proactively facilitating the participation of SMEs that would allegedly lack credit. In short, the Programme also takes a common practice – vulnerable SMEs need more credit guarantees for borrowing – into consideration when it comes to designing the financial structure of the Programme. In addition, the success of pilot project guaranteed by the Programme would prove the feasibility of the energy efficiency projects to the private financiers and local industry. It will create a sustainable energy finance ecosystem that can be financed on its own without the support from GCF and other external financiers, coupled with the impacts of TA components, such as de-risking mechanisms development.

Target Technology Selection to Maximise Economic Benefits. The Programme is also designed to take into account the economic benefits considering increased capital expenditures (CAPEX) due to technology and equipment installation and reduced operating expenses (OPEX) due to energy savings. The shortlisted technologies presented in the eligibility criteria were selected to maximise energy savings per CAPEX or another expenses, based on the information of each technology obtained from energy experts' group, including ACE (See Annex 2 and 3). From this perspective, it is deemed that the Programme is designed to produce the most substantial environmental benefit at optimal investment cost, given that the energy saving rate is closely connected to the rate of GHG emission reduction (See Annex 22).

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
☐ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	There are few cases of successful energy efficiency projects in Indonesia in the industrial sector, due to the government subsidised electricity tariffs and high upfront costs for energy efficiency technologies. The aforementioned reasons result in minimal participation of industrial players and other new entrants into the energy efficiency programme.	<u>Low</u>	Change will begin with industrial players' confidence that energy efficiency projects are no longer upfront cost-intensive, but rather attractive businesses offering higher operating profit, long-term productivity improvement, and more sustainable production processes. This will be achieved with the execution of enough successful energy efficiency projects at scale.	The intervention is projected to catalyse scalability of the industrial energy efficiency business area overall, by dissolving chronic financial, regulatory, and technical barriers through the de-risking guarantee facility and enabling market creation efforts which are currently lacking in the market at present.

Replicability	In that Indonesia is a market leader in ASEAN member states in various areas, the successful establishment of an energy efficiency ecosystem in this leading emerging market can be serve as a representative case to be replicated in other ASEAN member states. However, due to the current lack of successful cases, replication is not yet possible.	<u>Low</u>	Once profitable energy efficiency projects with adequate GHG emission reduction impact can be demonstrated, the associated financial and business structures will be naturally replicated to neighbouring countries that shares the similarities within the ASEAN community. In particular, the development of de-risking techniques, such as energy saving insurance (ESI), will encourage private investors to promote the energy efficiency business.	Lessons learnt through this Programme will directly impact and provide a pioneer example to any effort to replicate the work in other countries under knowledge sharing activities amongst ASEAN countries (Component 3). Positive perceptions of energy efficiency finance and de-risking mechanisms will spread across not only Indonesia but also across the ASEAN region over the long term.
Sustainability	A sustainable energy efficiency ecosystem cannot be achieved solely through enhanced access to finance. Considering the current regulatory barriers such as the subsidised energy tariff and lack of fiscal/non-fiscal incentives and the credit risk mitigation schemes that local financial institutions have implemented, the Programme sustainability remains in question even if several successful energy efficiency projects are implemented through the support of the GCF.	<u>Low</u>	Sustainability can be reached with a stronger regulatory framework to support the Programme's financing commitment. In addition to changes in the government's policies, the lack of awareness of energy efficiency technologies, capacities, and resources of industry players and external advisers need to be addressed in order for the prioritisation of investment in energy efficiency projects. Packaged efforts in building the regulatory and technical capacity of Indonesia will ensure the long-term sustainability of the energy efficiency financing programme, in particular, amid the COVID-19 pandemic.	The NDA-initiated Programme will work closely with government agencies and public sector financiers to build their financing capacity and refine the existing regulatory framework. The Programme will also provide capacity building activities, such as technical advisory services, to industry players, including financial institutions to be equipped with energy efficiency business pipelines and sustainability guidelines. The Programme efforts will result in the vibrant energy efficiency market creation and development that ensures sustainability, even far beyond the Programme implementation period and areas.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

*Please be noted that figures below are indicative as impacts vary depending on finally chosen industries and technologies in each bundle.

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ²³	
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	1. Programme GHG (Green House Gas) emission reduction reports conducted and verified by an independent verifier 2. Annual performance report including the respective GHG emission reduction from end borrowers and FIs	0 tCO ₂ eq	1.10 MtCO ₂ eq	2.825 MtCO ₂ eq	Key assumptions used in Annex 22 and based on the implementation period of 10 years. Note that the Programme total expected mitigation outcome of 3.139 MtCO₂eq is based on the technologies lifetime and disbursement schedule, which is 12 years.
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Supplementary 1.1: Annual energy savings</u>	1. . Programme energy saving reports conducted and verified by an independent verifier 2. Annual performance report including energy usages and savings,	0 GWh	3,919GWh	10,065GWh	Key assumptions used in Annex 3 and Annex 22

²³ The final target means the target at the end of project/programme implementation period (10 years).

		from end borrowers and FIs				
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	Project financial report on total cost and asset value from end borrowers and FIs	0 USD	USD142.3million	USD142.3million	Key assumptions used in Annex 3 and Annex 22

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	A weakly regulated institution of Indonesia discourages the promotion of the industrial energy efficiency; institutional barriers include the low price of electricity by the distorted government subsidies, no regulation on energy efficiency specific areas, and no market standards such as MEPS.	<u>low</u>	Ministries and public agencies with their own varying insights will closely coordinate to formulate the proper desirable regulatory framework with policy incentives. Once accepted, Indonesia will have an enabling institution attractive enough for relevant stakeholders to join the industrial energy efficiency market.	Relevant local stakeholders will be trained, and their capacity will be built with the Programme's policy and regulatory intervention, which will result in reasonable tariff adjustments and enabling institutions for energy efficiency promotion in Indonesia.	<u>National level (one country)</u>

Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Market players have a low prioritisation of investing in the energy efficiency business, due to many barriers such as no recognition on regulatory requirements and little confidence from the economic perspective on business.	<u>low</u>	Indonesia will witness the formulation and facilitation of a self-funding market where market players could function with the protective de-risking mechanisms newly deployed in the market.	An enabling energy efficiency market will be created and developed with the key market players participating in the Programme's energy efficiency finance under Component 1.	<u>National level (one country)</u>
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Potential players lack awareness of the industrial energy efficiency area and relevant regulations, while there exist a few well-trained/experienced energy experts amid no market incentives for employing them.	<u>low</u>	Indonesia with several best practices and enhanced capacity in energy efficiency finance will spread and share newly accumulated lessons nationwide and region wide.	Effective knowledge generation, awareness enhancement, and learning processes will be achieved with use of best practices and de-risking methodologies and standards under Components 2 and 3.	<u>Multi-countries</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "Project/programme co-benefit indicators" section in table below. Add rows as needed.

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Increased volume of energy efficiency financing in Indonesia industry with an increased participation of the LFIs in energy efficiency financing	Number of LFIs participated in the Programme	Financial contract with KDB	0	6	6	LFIs who are not Executing Entity in the programme are not counted
	Number of projects approved by local FIs	Annual reporting to KDB from local FIs (if needed, the frequency of reporting may increase)	0	20 (30% for women-led/owned SMEs)	20 (30% for women-led/owned SMEs)	# of projects is subject to change as per local needs GAP
	USD Volume of disbursement – both lending of final beneficiaries(sub-project loan) and credit guarantee of the GCF	Semi-annual reporting to KDB from local FIs (if needed, the frequency of reporting may increase)	0	(Sub-project loans) 133 mil USD (guarantee) 100 mil USD	(Sub-project loans) 133 mil USD (guarantee) 100 mil USD	As per disbursement under each sub-project agreement
Generation of locally customised ESI/ESCO financing mechanisms	Documented ESI / ESCO schemes for sharing with local players	ESI / ESCO document approved by local FIs and local enterprises, with annual reporting to KDB	0	1	1	The document should be evaluated and approved by KDB (or independent advisor)
Generation of locally customised ESI/ESCO financing mechanisms	Strengthened financing capacity of local FIs	Documented financing capacity building manual	0	4	4	As for 'documented financing capacity building manual', the number depends on the number of EEs (local FIs).
	Strengthened technical capacity of industry players	Documented technical capacity building manual per energy efficiency technology/equipment	0	5	5	
	Strengthened regulatory framework with policy incentive recommendations	Documented policy recommendations submitted to NDA and relevant ministries	0	1	1	All the activities should be annually reported to KDB
Spraeading positive perceptions of EE Finance in Indonesia and ASAEN	Accumulation of know-how and lessons learnt	Documented knowledge sharing material, submitted to KDB, local FIs, and NDA	0	0	5	Number depends on the number of participating countries. All the activities

						should be annually reported to KDB
	The number of pilot projects in ASEAN member states	Documented pre-feasibility studies of pilot projects	0	1	2	Number depends on the number of participating countries. Pre-feasibility should reflect lesson learnt from the
	Active knowledge sharing for ASEAN replication	Workshops with participants across ASEAN	0	0	5 (number of workshops), 100 (number of participants)	Workshop will be annually held since mid-term period, with best practices accumulated All the activities should be annually reported to KDB
Project/programme co-benefit indicators						
Economic Co-benefit: Economic benefits of industries overall with saved energy and jobs created accordingly	Industrial profits caused by saved energy costs	Converted value of energy savings	0	USD27million (annual average)	USD36million (annual average)	[Assumption] - Energy tariff of US\$ 0.078/kWh for businesses (medium voltage) ²⁴ / F/X Rate of '1 USD = 14,504 IDR ²⁵ - Other key assumptions used in Annex 3 and Annex 22 Target value is subject to change as per local context, depending on energy tariff and FX rate in Indonesia
	New energy efficiency jobs created	New employment in financed industrial companies, ESCO,	0	200	600	[Assumption] - Average annual salary of US\$28,927 ²⁶ , and 5% of

²⁴ <https://www.esdm.go.id/en/media-center/news-archives/electricity-tariffs-for-april-june-2021-remain-unchanged>

²⁵ As of 7 April 2021

²⁶ Mechanical Engineer, 2021, in Jakarta (www.teleport.org)

		relevant energy efficient technology areas				additional industrial profits is used to hire New energy efficiency jobs - Other key assumptions used in Annex 3 and Annex 22 More specific details will be reported to AE annually.
Gender Co-benefit: Women's active participation in the energy efficiency labour market, with an increase of skilled female engineers	Increased women's access to energy efficiency finance	Ratio of women-led/owned SMEs out of all borrowers	0 ²⁷	30% of financed beneficiaries	30% of financed beneficiaries	As per Gender Action Plan
	Gendered stakeholder consultations as per sub-project approved by local FIs	Participation of women amongst stakeholders	0 ²⁸	20% out of consulted participants as per project	20% out of consulted participants as per project	Gendered stakeholder should embrace male stakeholders, and encourage them with a gendered mindset
	Gender capacity building trainings	The number of workshops and/or seminars	0	5	10	At least one gender event as per year

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity 1. Backstopping credit risk of local FIs' own loan with the GCF credit guarantee provision	When a FI would provide an end borrower with its own loan under pre-agreed eligibility criteria, it would be entitled to benefit from GCF	1. FI's energy efficiency project selection as per eligibility criteria 2. Guarantee proceeds disbursed	1. GCF Guarantee Cover of USD 100 million disbursed

²⁷, Baseline is zero because Indonesia rarely have energy efficiency finance disbursements successfully done so far; a few were identified non-gender mainstreamed as a result of the local FIs consultation.

²⁸, Baseline is zero because Indonesia rarely have energy efficiency finance disbursements successfully done so far; a few were identified non-gender mainstreamed as a result of the local FIs consultation.

	Guarantee Cover to be provided by GCF (via KDB) against credit risk of the end borrower up to a pre-agreed guarantee coverage ratio. Guarantee Advisor employed by KDB will monitor and manage the guarantee facility with a focus on F/X risk management.	1. Monitoring and follow up management by Guarantee Advisor employed 2. F/X risk management	2. FI own loan of USD133 million disbursed
Activity 2.1: Introduction of customised Energy Saving Insurance (ESI)	Developing de-risking mechanism of an energy saving insurance model and scheme that would help lower a high level of perceived risk perception, and would establish credibility and comfort amongst key relevant actors assessing priority energy efficiency technologies to be de-risked through the market study.	1. Develop locally customised i) insurance covering energy savings, ii) standard ESI contracts, iii) energy savings validation methodologies 2. A Consulting and in-depth interview with key stakeholder and technology providers. 3. Capacity building and workshops	1. Report on locally customised de-risking mechanisms including standard ESI contract its and validation methodologies (approved by AE)
Activity 2.2: Development of ESCO financing structure	ESCOs will be introduced with a standard format, such as outcome-fixed/guaranteed contracts and capacity building and awareness enhancement.	1. Develop a locally fitting outcome-fixed contract 2. Capacity building and workshops for awareness enhancement and the ESCO implementation	2. Report on locally fitting ESCO financing structure
Activity 3.1: Local FIs capacity building activities	Supporting the local FIs to be equipped with the energy efficiency business pipeline in compliance with a set of eligibility criteria, with operation capacity of implementation the Programme in line with sustainability principles such as E&S and gender safeguards	1. Support for the development of project pipeline including seminar or workshops to industry 2. Technical (E&S, gender) advisory on monitoring and evaluation on projects (for Local FIs)	Project pipeline report/Technical advisory submitted to KDB
Activity 3.2: Technical advisory for industry player	Supporting the local industry stakeholders to be equipped with the energy efficiency business execution capacity	1. Technical advisory on the technology installation or utilisation for industry players 2. Technical (E&S, gender) advisory	Technical advisory book for local industry stakeholders with capacity building workshops

Activity 3.3: Development of energy efficiency regulatory framework	Supporting Indonesia to fine tune, strengthen, and document the regulatory framework on the energy efficiency investment and technology as a result of review and a series of consultation with Indonesian policy makers.	1. Capacity building and workshops with local policy makers with consulting and in-depth interview with key stakeholders 2. Documentation of policy guidelines and regulatory framework for recommendations	Regulatory framework and Policy guidelines approved by AE and submitted to NDA and relevant ministries
Activity 3.4: Sharing knowledge and understanding gained by Programme with ASEAN member states	Support the regional scale up of the Programme throughout ASEAN member states by extracting the lessons learnt and understanding gained by experiences in Components 1 and 2 and share it with countries in the ASEAN region.	1. Draw the lessons learnt from the Programme implementation 2. Knowledge sharing session with key stakeholders in ASEAN through workshops or webinars	Capacity Building / Knowledge Sharing Book submitted to AE

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

The Programme plans monitoring, evaluation, and reporting activities of all components in a strategic way. As there will be a need for closer monitoring and communication to ensure a sound start of the Programme, more frequent arrangements will be scheduled during the initial stages of the Programme. Monitoring and reporting activities will be arranged to begin on a quarterly basis and move to semi-annually as the development pipeline and Programme matures. In order to avoid any confusion, the interim and final independent evaluations will be conducted in accordance with the GCF's evaluation criteria as per the GCF Evaluation Policy

Component 1

Since the programme includes various levels of stakeholders such as final beneficiaries and financial intermediaries, it would be important to determine the appropriate channels for monitoring and reporting in accordance to the accountability of each stakeholder.

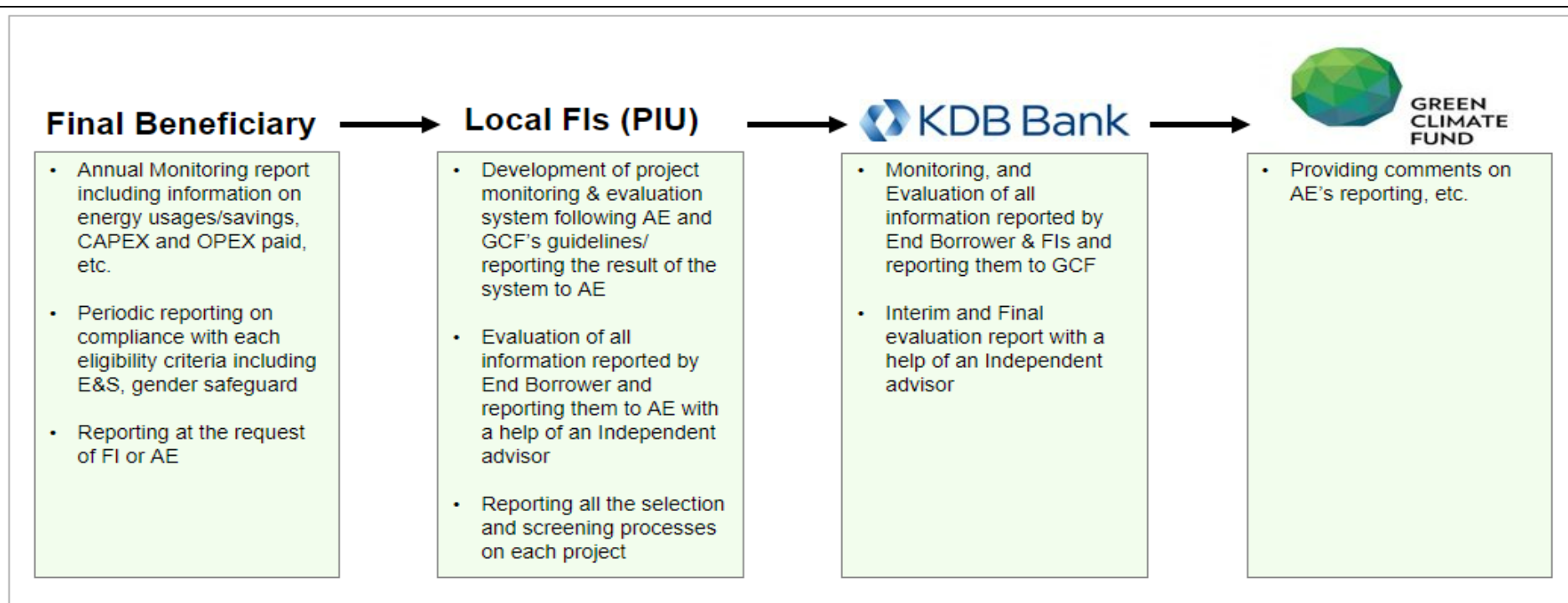
First of all, it is anticipated that the final beneficiaries (the borrowers) will report the financials of installing, setting-up and operating the equipment as well as the amount of energy consumed. This data will be used to determine the energy savings of each equipment installed. The borrower will have to ensure that they report the correct type of fuel such as electricity, and the correct input data using identical units for the reporting period. This data will be reported to the executing entities (local financial intermediaries).

The LFIs will then gather all information from their final beneficiaries and check the figures for both energy consumption level and the corresponding total emission savings. After an initial check, all data will be streamlined and reported to both KDB(AE). The summary of all data will be organized and confirmed. After this stage, KDB will report to GCF regarding the programme's overall energy and emission savings level in tandem with the financial aspects. This whole process outlined above will be discussed and confirmed once the programme has been initiated. In

addition, KDB and local intermediaries will endeavour to collaborate so that data is monitored at an appropriate level alongside verification of certain sample data to ensure data reliability and validity.

- **Final Beneficiary:** Final Beneficiary is the entity that performs an actual energy efficiency project and is obliged to report basic data, such as energy usage/savings and expenses/cost for project monitoring and evaluation. However, due to the nature of the Programme that focuses on SMEs, the vulnerable group of final beneficiaries has realistic barriers and difficulties in establishing a separate business unit for each energy efficiency project or having expertise for project monitoring activities. Therefore, the main obligation of monitoring and reporting to the AE and GCF including information interpretation, impact and assessment besides the basic information reporting is transferred to local FIs under Sub-project Loan Agreement.
- **Local FIs:** FI is a local financial institution selected by the AE, as it has provisional track records of high efficiency, effective monitoring, and supervising of country-wide programmes that has the capacity for monitoring and evaluation to support this type of project. However, their experiences with GCF's E&S/gender guidelines, or energy efficiency project may be not sufficient to fulfil the necessary guidelines. Against this backdrop, a Programme Implementation Unit (PIU) will be formed within the local FIs that will be in charge of monitoring, reporting, and evaluation duties. The PIU will be composed of GCF project managers, focal points, and consultants dispatched from external expert agencies or TA Providers under Component 3 of the Programme. Potential external expert agencies are shortlisted in the programme development stage through local stakeholder consultations. Other TA activities under Component 3 support the development of pipeline, technical, E&S monitoring, and evaluation activities of FIs.
- **KDB (AE):** KDB monitors all information reported by local FIs on a quarterly basis (potentially shift to semi-annually as the Programme matures), evaluates and assesses whether each project is in line with the GCF and KDB guidelines, and provides the supervisory function for the Programme lifespan. KDB also communicates with the GCF throughout the procedures of interim and final evaluation reporting to the GCF. All reporting results will be faithfully delivered to the GCF with assurance of the proper business operation throughout the Programme lifecycle. KDB will have meetings, at least quarterly with each of FIs for more efficient and active project monitoring and evaluation activities.

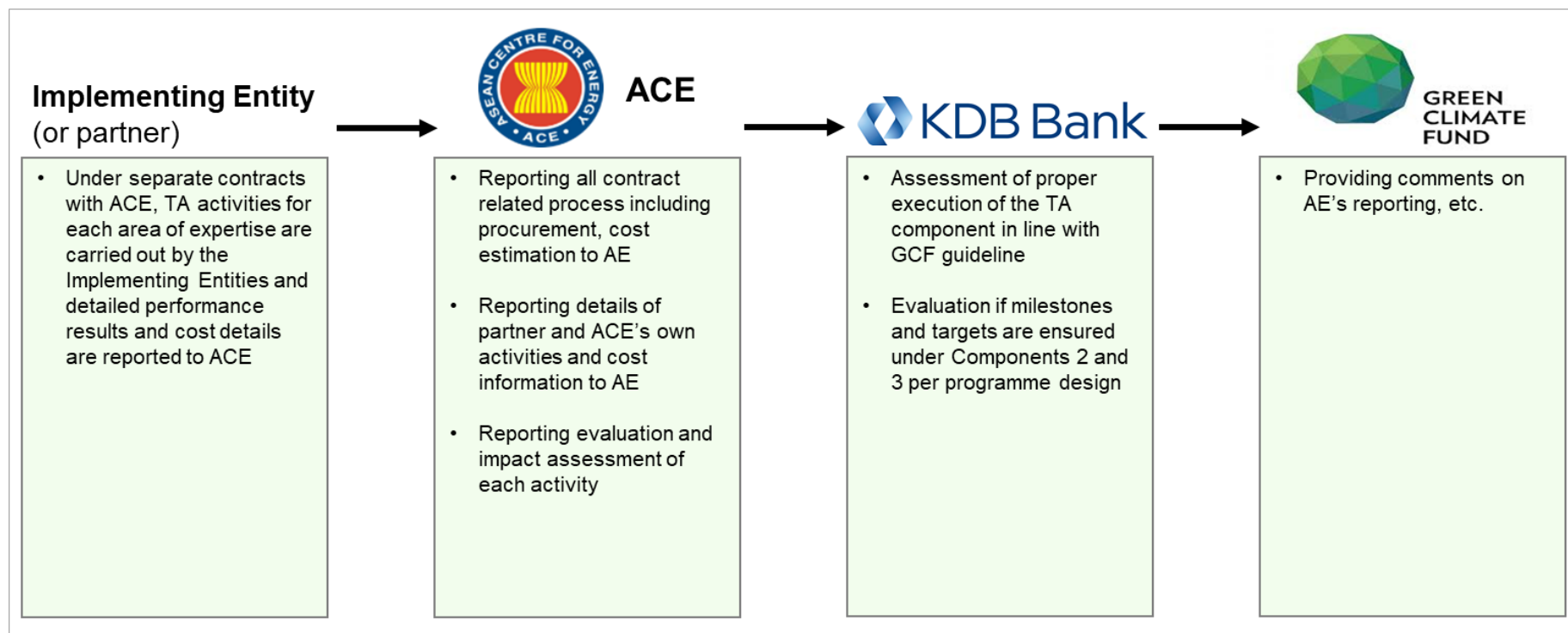
[Figure 19] Monitoring, reporting and evaluation arrangement under Component 1



Components 2 and 3

Monitoring and evaluation on TA activities under Components 2 and 3 are to be conducted by ACE on the ground. ACE assesses and reports whether the selection of Implementation Entity (IE) and contracts with the respective IE is in line with GCF and KDB guidelines. ACE will receive reporting on detailed information about TA activities and cost information from each IE on a quarterly basis (frequency will be re-evaluated after Programme initiation). In addition, ACE will analyse if TA activities have been properly implemented as planned, evaluate their effectiveness, and report its results to KDB on a quarterly basis. KDB will finally review all the reported information, confirm the proper execution of TA activities, and reports its results to the GCF quarterly, initially and potentially semi-annually as the Programme matures.

【Figure 20】 Monitoring, reporting and evaluation arrangement under Components 2 and 3



Leveraging Local Capacity amid the COVID-19 Crisis. A strong partnership with local expert entities under the financial intermediation system is important, particularly, in the midst of the global pandemic outbreak. COVID-19 has begun to rapidly spread within a country and even from country to country around the globe with rising infection rates and loss of life. Faced with this uncontrolled stage of the COVID-19 with a lagging roll-out of vaccines, KDB cannot directly visit there and keep an eye on project performance and compliance with requirements due to an overseas travel ban, but ought to necessarily depend on reporting outcomes of local partners. Therefore, the Programme undertakes a form of periodic monitoring, reporting, and evaluation on the basis of systemic collaborations with executing entities and partners of the Programme, including Indonesia NDA. As an accredited entity, however, KDB shall come up with an idea of effective supervision and shall increase the frequency of periodic site visits as the number of infection cases gradually go down.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		

A potential technical and operation risk might be caused by a lack of knowledge and competence of end beneficiaries – mainly industrial enterprise owners – surrounded by the least-developed energy efficiency market circumstances. Especially, industrial enterprises ought to adopt themselves to change into energy efficient equipment and technologies. The change necessarily requires enterprises to redefine the operation and management (O&M) process as per modification and/or replacement of technical methodology and operational mechanism.

Mitigation Measure(s)

The eligibility criteria require local FIs and their borrowers (end beneficiaries) to carry out a technical assessment on underlying projects so that any potential technical and operational risk attributable to unfamiliarity and a lack of skill and knowledge could be minimised as much as possible. The set criteria further require demonstration of a concrete O&M plan over the borrowing lifespan in order to ensure the energy efficiency system optimisation and energy saving results accordingly. As per request from partner FIs, the Programme provides support for a technical assessment of the vulnerable group of local beneficiaries under Component 3.

Selected Risk Factor 2

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>
Description		

A potential governance risk might be caused by the Programme implementation governance of a standard guarantee intermediation structure under which KDB relies on local FIs with a full authority to select, appraise, structure, execute, lend, and manage sub-projects and their proponents. Given that KDB controls on local borrowers operate only via the FIs, the selected FIs are the key to implement the Programme faithfully in line with the eligibility criteria, for instance, including ESS and the gender lens investing principle. This would relate the classic principal-agent problem often arising due to inherently competing priorities between the principal (GCF and KDB) and the agent (local FIs).

Mitigation Measure(s)

Partner FIs selection has been cautiously done with many rounds of discussions, including a firm institutional commitment to the GCF and KDB policies and guidelines. In addition, a partial guarantee coverage arrangement (100% coverage not acceptable as per eligibility criteria) under Component 1 is a protective device to align FIs' interest with that of the GCF and KDB, thereby minimising the chronic principal-agent issue. On top of the above, the Programme puts multi-level contractual and institutional arrangements to be effective throughout all the business cycle in place: (1) the eligibility criteria for sub-project selection; (2) the Sub-Project Loan Agreement aligned with the FAA; and (3) the operation of the PIU. Also, KDB will monitor FIs, through reporting and physical visits where possible, if FIs implement the Programme in line with the eligibility criteria, and the guidelines of KDB and GCF, which therefore expects periodic meetings between KDB and each of FIs, including the Guarantee Advisor.

Selected Risk Factor 3

Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>

Description

The GCF Guarantee Facility of Component 1 would be directly exposed to the risk by its nature, up to a pre-agreed guarantee coverage ratio. As typically seen in a credit guarantee or insurance product, the guarantor could be also exposed to any credit risk attributable to information-asymmetry.

Mitigation Measure(s)

Shortlisting and selection of local FIs takes the credit rating and relevant reputation into account. In particular, SMI and IIF, under the current discussion, are reputable private financial institutions in Indonesia and their respective credit ratings are equivalent or close to sovereign rating of the country. As such, it does not foresee the credit risk to be potentially material.

As for the FI Own Loan to be backed by the GCF guarantee Cover, the FI would share any credit risk of the relevant borrowers up to the residual portion under the guaranteed FI Own Loan. With the said mechanism, any credit risk of the final beneficiary exposed to relevant FI is not insignificant, thus the moral hazard risk to be less likely to occur. What is more, equity injection from a final beneficiary could be told as another mitigation measure against any credit risk of borrowers.

Selected Risk Factor 4

Category	Probability	Impact
Forex	Low	High

Description

There is a F/X risk of the Indonesian Rupiah (IDR) against USD because the vast majority of FI Own Loan to final beneficiaries are likely to be provided in IDR as it is the case for a typical corporate loan to domestic borrowers; however, GCF funding (Guarantee) is provided only in USD. As the GCF Credit Guarantee is denominated and made available in USD, FIs (as guarantee recipients) may be exposed to a currency mismatch risk related to the currency of defaulted debt service amounts under the FI Own Loan (in IDR, but this currency mismatch risk is limited to the case where IDR appreciates against USD).

Mitigation Measure(s)

The GCF Credit Guarantee will be designed in a way which the foregoing F/X risk would be partially absorbed by the GCF and F/X risk not absorbed by GCF is borne by local FIs. F/X risk will be regularly monitored and managed by a professional Guarantee Advisor who would be employed by KDB.

Selected Risk Factor 5

Category	Probability	Impact
Legal	Low	Medium
Description		

The programme entails multi-dimensional and multi-level legal and contractual arrangements with multiple parties, which might carry a potential legal risk in association with legality or validity and enforceability of any agreement or contract. Furthermore, dis-alignment amongst agreements and contracts governed by different jurisdictions might be another potential legal risk.

Mitigation Measure(s)

In order to avoid a potential legal risk arisen by jurisdiction discrepancy, definitive agreements where KDB is a contracting party – i.e., FAA, Guarantee Agreement, and TA Service Agreement – are to be universally governed under English Law and potent disputes are to be resolved by arbitration administered by the Singapore International Arbitration Centre (SIAC) in accordance with the SIAC Arbitration Rules. This is to be incorporated in the agreements.

On the other hand, all the Sub-Project Loan Agreements to be entered between a FI and each borrower are likely to be governed by the Indonesian law – the host country law. That implies that all the sub-transactions are standard corporate loan financing, which are the core business areas of local FIs with an extensive track record over decades. In that regard, the occurrence of any legal risk, potentially caused by jurisdictional discrepancy seems marginal; and even if a potential risk occurs, local FIs are deemed capable enough to tackle the legal disputes by themselves as a contracting party versed in the local legal process.

In addition, all legal agreements in the programme will be carried out with support and advice from qualified independent legal advisors, which will minimize the legal issues and controversies.

Selected Risk Factor 6

Category	Probability	Impact
ML/FT	Low	High
Description		

The GCF proceeds under the programme can also be utilised to support anti-terrorist activities, money laundering or prohibited activities policies by FIs or the final beneficiaries. Additionally, this risk of a failure to meet international standards with respect to preventing money laundering and/or financing of terrorism can present problems, such as obstacles to financial flows and/or implementation and/or payment processing.

Mitigation Measure(s)

The programme under Component 1 is supporting credit guarantees with local FIs, who will provide FI Own Loans to the final beneficiaries. As FIs were or will be selected through the proper capacity assessment by KDB including AML/CFT (KYC) assessment, final beneficiaries must also undergo due diligence according to the Sub-Project Loan Agreement between the FI and the final beneficiary. Then, it is considered that such related risk has been substantially mitigated. Also, Indonesia has the national regulatory system to monitor relevant issues of the financial institutions under the purview of central bank. Moreover, the potential Executing Entity under Component 1 has an internal regulation and policy about their investment activities, so the final beneficiaries' activities under the programme will be properly screened and monitored.

Under Components 2 and 3, AML/CFT assessment on the Executing Entity (ACE) is already completed by KDB, and the activities of implementing entities will be monitored if they are in line with the ACE's internal regulations agreed by KDB. Reporting of complaints and allegations of impropriety, wrongdoing etc., are conducted by PIU (Programme Implementation Unit) for Business Coordination and Sustainability under KDB Supervision, which will be established in each local FI under Component 3.

Selected Risk Factor 7

Category	Probability	Impact
Sanctions	Low	Medium

Description

There should not be any United Nations Security Council resolutions imposing financial sanctions that would prohibit the engagement of GCF or the AE in the country or with individuals or entities under each Component of the programme. If there is any, any approvals, exemptions, licences, or other permissions are required.

Mitigation Measure(s)

KDB has similar measures announced by the United Nations (UN) or other international organisations (e.g., World Bank, OECD, IMF, etc.), and will not deal with the countries or companies included in the list. In addition, it was confirmed by the internal system that Indonesia and ACE are not included in the list as of now.

Selected Risk Factor 8

Category	Probability	Impact
Prohibited practices	Low	Medium

Description

There should not be any practice included in GCF and KDB prohibited practices lists, e.g., non-compliance with the E&S safeguards and gender discrimination risk as for employment, during the implementation and financing of the activities, under the GCF and KDB guidelines on prohibited practices.

Mitigation Measure(s)

KDB has an internal system that controls AML/CFT and other prohibited practices, including prohibitions against E&S and Gender mainstreaming standards, and all the activities under the Programme will be monitored and supervised by KDB if any of them is in the list, based on GCF and KDB guidelines.

Selected Risk Factor 9

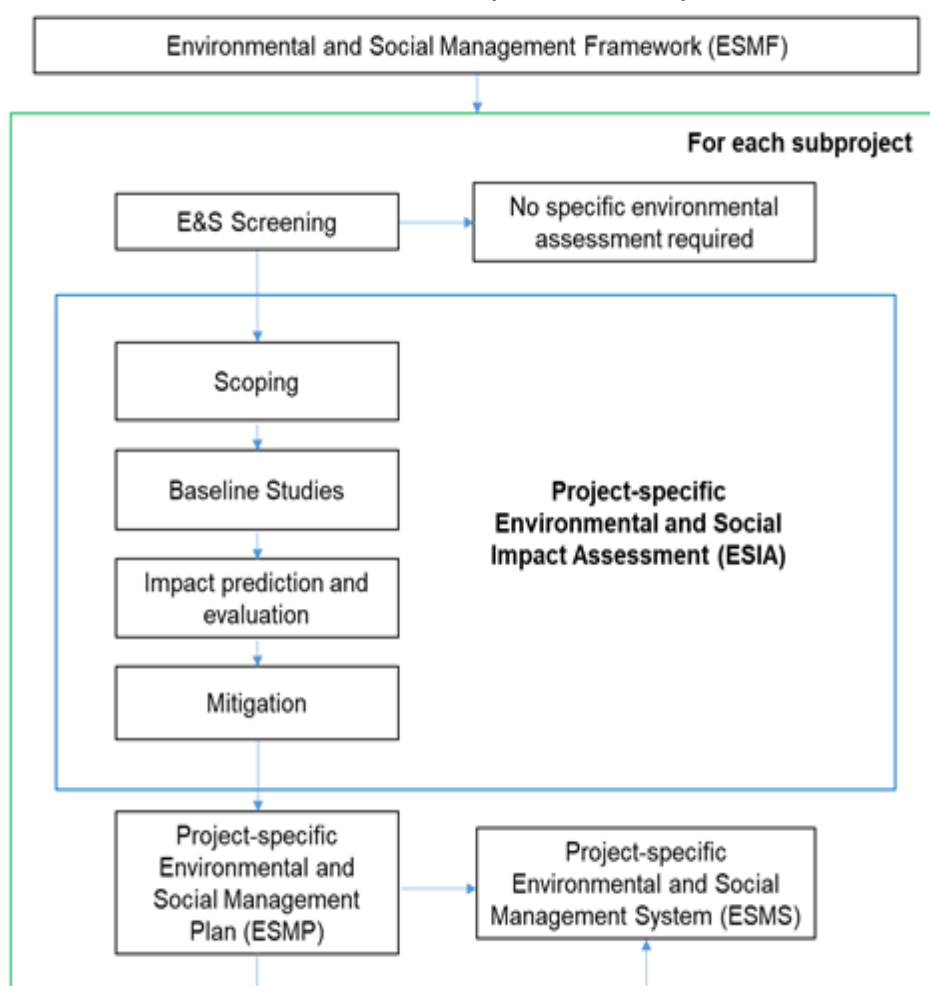
Category	Probability	Impact
Regulatory Risk	Medium	Low
Description		
Regulatory barriers including lack of fiscal and non-fiscal incentives, fossil fuelled energy subsidies, etc. should be addressed for energy efficiency improvement in Indonesian industrial sector from a long term point of view. However, there is a risk that it will not improve to a satisfactory level, given that it needs to coordinate various opinions or interests between many kinds of government ministries and there may face some conflict with the economic development rationale which is one of the government's main policy goals.		
Mitigation Measure(s)		
The Programme is aiming to minimize this risk through Activity 3.3 under Component 3, In addition, it has a structure that enables a successful programme implementation without a prompt improvement of policy and regulation through GCF guarantee Cover and de-risking scheme development. Also, if energy efficiency project is successfully implemented and its potential economic benefit is known to policy makers, the government's willingness to improve the energy efficiency regulatory framework is expected to increase.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Environmental and Social Management System (ESMS). The Programme submits an ESMS as Annex 6 of the funding proposal package for the integration of Environmental and Social (E&S) aspects into the energy efficiency initiative, estimating the E&S risk category B. Given that project-specific information such as location and project size is unknown at a programme level, the ESMS has been developed to provide guidance for screening and assessing likely E&S issues, carrying out individual project-specific environmental and social impact assessments (ESIA), and preparing the environmental and social management plan (ESMP) to mitigate potentially negative E&S risks. The framework will support local FIs and their borrowers to ensure compliance with the GCF and KDB safeguards policies and the government policies, rules, regulations, and procedures.

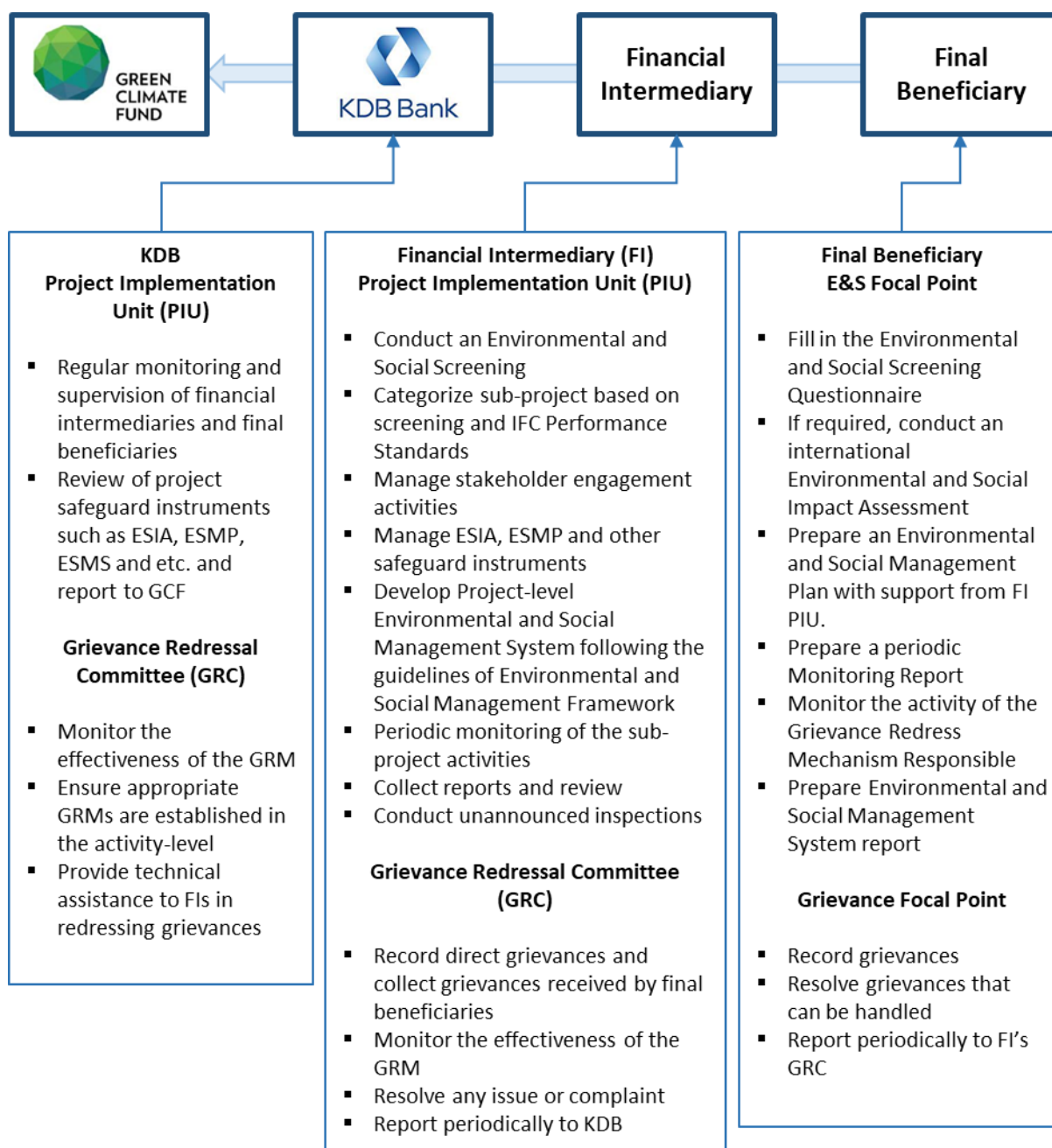
【Figure 21】 ESMS with E&S Framework, Assessment, and Plans



Potential E&S Impacts and Proposed Mitigation Measures on the Basis of a Phased-Approach. The framework singles out the followings as potential E&S impacts; thermal energy used by factory, electricity used by factory, surface and ground water pollution, air and dust pollution, noise pollution, solid and hazardous waste management, and suppliers' compliance with labour and environmental standards. They are arranged in three phases – pre-installation, installation, and post-installation phases – with mitigation and control measures and responsible entities. Please see Appendix E of the ESMS for more details.

Institutional Arrangement for ESS Compliance. A robust institutional arrangement should be present in order to effectively comply with stipulations given under the framework. Under a two-step lending structure, each of the local FIs will have its own PIU which will be partly composed of the E&S expert group. As an intermediary, local FI is responsible for periodic monitoring and reporting to KDB on sub-projects they finance, and these duties are to be carried out by the PIU. The programme will be in compliance with the GCF Policy on the Prevention and Protection from Sexual Exploitation, Sexual Abuse, and Sexual Harassment (SEAH Policy) as prescribed in its ESMS (Annex 6). According to the SEAH Policy, the programme sets clear obligations for all individuals in the GCF funded activities to prevent and respond to the SEAH, and to refrain from condoning, encouraging, participating in, and/or engaging in SEAH. This will be also addressed by the PIUs across local FIs.

【Figure 22】 Implementation Arrangement for E&S Compliance



Consultations with Key Stakeholders – NDA, Relevant Policy Makers, CSOs, etc. – and Potential Impacts on Indigenous Peoples. For gathering practical comments on the group, the Programme has also undertaken a governmental level focus group discussion with GCF, NDA, OJK, Fiscal Policy Agency (Badan Kebijakan Fiskal, BKF), PLN, MENR, and ACE, and in-depth consultations with civil and society organisations (CSOs, GCF observers). Part of those consultations revealed the likelihood of potential impacts on indigenous communities; for instance, a CSO - WALHI, Friends of the Earth Indonesia - shared an expert view that pulp and paper industries have been using forestry resources of indigenous people, and that cement companies have affected water conservation areas which may give impacts to biodiversity and indigenous community livelihoods. The Programme will not finance any project with potential adverse impacts on indigenous people. Please see below a summary of the CSOs representing Indonesia and refer to the ESMS for more details including identification and selection of interviewers and minutes of interviews.

【Table 14】 Summary of Consultations with CSOs

No	Stakeholders Engaged	Date	Summary
1	IESR (Institute for Essential Service Reform)	Friday, 26 March 2021	There are many energy efficiency projects implemented so lessons learned should be adopted. Opportunities to do energy efficiency are seen a lot in cement industries and commercial office building. Each project under the Programme has to submit an environment and social impact assessment suggested by the GCF ESS.
2	KEHATI (Yayasan Keanekaragaman Hayati Indonesia - Indonesia Biodiversity Foundation)	Friday, 19 March 2021	The Programme implementation should highlight the ability of the Programme to reduce carbon footprint. Human right or gender aspect have not seen as one of the noticeable components of project implementation in energy efficiency.
3	WALHI (Wahana Lingkungan Hidup – Friends of the Earth Indonesia)	Wednesday, 24 March 2021	WALHI suggested that each programme related to energy efficiency should think about the impact that affects the environment and community surrounding the company facilities or down the company's supply chain. Some examples are that pulp and paper companies have been using Indigenous People's forestry resources. Mining companies (cement) has destroyed water conservation area which may impact biodiversity and community livelihood. The textile and food industry has been known to have a problematic approach to their workers' policy. Stakeholder consultation with the observer should not only happen at the beginning of the program designed only. The final draft of the Programme /screening instrument should allow a discussion with observers as well.
4	Perserikatan Solidaritas Perempuan ((SP) Women's Solidarity for Human Rights)	Friday, 26 March 2021	SP suggested adopting SP Guidelines for Women Safeguard on Policy, Program, and Climate Change Project.

Grievance Redress Mechanism (GRM) with Grievance Redress Committee and Focal Point. The programme sets out the GRM that answers queries, receives suggestions, and addresses complaints and grievances about any irregularity in the application of the guidelines adopted in the ESMS. As an executing entity and a financial intermediary, local FIs are to form a Grievance Redress Committee (GRC) and designate a GRM focal point in their PIUs. As for the GRM process, all complaints and suggestions of aggrieved people will be formally received in a written form (Grievance Redressal Form in Appendix H of the ESMS) by a GRM focal point for a prompt response and the GRC is responsible for a periodic reporting to KDB while examining documentary evidence submitted during informal hearings and recording salient points to be presented by the aggrieved.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Gender Mainstreamed since the Programme Inception. Since making a scratch outline of the concept, a gender specialist has engaged the Programme development for the sake of a genuine gender mainstreamed work on the basis of KDB internal regulation. In that, the energy efficiency financing initiative differentiates

itself from the general grant projects. The gender studies have a strategic focus on financial inclusion of the vulnerable female group in terms of access to financial products and services; they are generally marginalised due to double or triple vulnerability like locality, low literacy, and lack of tertiary educational attainment as well as gender-caused lack of access to finance. Considering a serious lack of first level micro / raw gender-disaggregated data, literature review, focus group discussion, and key informant interview were used as the primary methodologies for data collection and assessment. A full-fledged gender document (assessment and action plan) is submitted as part of the funding proposal package.

Enjoying Market Opportunities through Gender Mainstreaming. The Programme sees the group of women-led SMEs as a market opportunity, rather than risk, with untapped potential in that the group makes up almost half of the country's SME sector. Despite unavailability of gender-disaggregated microdata and precise statistics on women's economic contribution, IFC (2016) estimates that 43% of formal SMEs are women-owned in Indonesia; informality is one of issues from a gender point of view though.²⁹ It accordingly mainstreams gender considerations during the lifespan, believing that serving women-led SMEs eventually contributes to the whole economy.

Supporting Underbanked Women-led SMEs through a Gender Lens Investing Approach. Above all things, the Programme shall guide and support local FIs to adjust their energy efficiency finance to better serve women-led SMEs. Most female entrepreneurs face inherent challenges and difficulties – e.g., collateral requirement and complex loan application procedures – in obtaining lending services from formal financial institutions for their business; the same might be true of beneficial energy efficiency financing. In order to tackle the harsh reality, the Programme produced the gender document (assessment and action plan) that applies the women-led SME framework under the financing structure: i.e., priority given to relevant women-led SMEs in terms of access to energy efficiency financial products and services.

In order to minimise anticipated discrepancy between studies and implementation, the Programme emphasises the practical and realistic solutions of the gender lens investing approach. For a practical use of the women-led SME framework, the Programme clearly defines a women-led SME as a formally registered enterprise under 250 employees (with reference to the United Nations definition on the medium enterprise), where at least one owner is a woman or at least half of employees are women. Gender action plan (GAP) sets out representative KPIs as follows; 20% of women-led SMEs: and 30% female participants in trainings, and women-only consultations.

Embracing Borrowers Not Ready Yet, Bringing Them towards a Gender Mainstreaming Commitment. Most importantly, the Programme does not discourage the borrowing business group not eligible for the gendered priority and financial benefit. It shall function as a meaningful incentive that encourages the not-ready group to take this opportunity to internally mainstreaming gender considerations: e.g., In the case of approaching local FIs with a commitment to employing more women, the committed business group will enjoy the benefits under guidance of the GAP. Consequently, the Programme aims to achieve Sustainable Development Goal 5 – gender equality and women's empowerment – and to produce best practices for the Indonesian SME banking sector as a whole.

Inputs Integrated from Consultation with Key Stakeholders Consultation. The Programme's gender document integrates valuable inputs as a result of a series of consultations with the Women's machinery and other line ministries and Indonesia-based key stakeholders with relevant expertise and a status as a GCF observer: Solidaritas Perempuan, IESR, Kehati, and WALHI. One piece of valuable feedback was a division between upstream and downstream energy efficiency business areas in analysing gender-biased vulnerabilities, which differentiates from other studies in taking a look at varying dynamics even within the energy efficiency sector. Detailed analyses and minutes of meetings are available in the submitted gender document. In addition, the Programme plans women-only consultations as prescribed in GAP to extract women's voice and call over the course of implementation.

²⁹ IFC (2016). Women-owned SMEs in Indonesia: A Golden Opportunity for Local Financial Institutions. March 2016.

Implementation Arrangement in Partnership with Local Gender Experts. As a Seoul-based accredited entity without a strong local presence, KDB has considered the most optimal implementation structure of partnering with local entities with contextual knowledge, relevant expertise, and the proper network and resources. That said, the consultation process was part of tapping into local agency partners to ensure the proper and sustainable implementation of planned activities including monitoring and evaluation, of course under KDB guidance and supervision. More on the local gender agency selection is to be discussed with Indonesia NDA and local FIs; the local FIs are at the centre of the women-led SME banking activities in actual and they will be supported by the local gender expert group. KDB will require selected local FIs to establish a PIU where local gender experts are to be assigned and to provide advisory on the ways to implement the GAP: e.g., what sort of priority to be given to women-led SMEs on the basis of banking operation, how to sort out them out of the pool of borrowers and screen them for final access to energy efficiency finance, and what specific activities and commitments to encourage the borrowers to adopt for mainstreaming gender considerations.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Systemic Governance and Monitoring Structure Ensures Proper Financial Management. The Programme's funding flows will be regularly monitored and strictly governed by a set of contractual and institutional arrangements under the guarantee facility implementation structure, as addressed in the section B.4. In principle, the GCF proceeds will be independently managed in a KDB account pursuant to the general provisions of the AMA and FAA. As the Accredited Entity, KDB, will regularly confirm whether financial operations, such as disbursement of the proceeds, are being properly executed in compliance with contractual and institutional arrangements.

As for the GCF Credit Guarantee proceeds, each of the Guarantee Agreements will make a reference to relevant covenants set out in the Guarantee Agreement while imposing additional undertakings and covenants customised for the guarantee product. By doing so, the guarantor and its agent will be fully informed of all of the material information over the guarantee execution period, so as to avoid any moral hazard issues related to potential information-asymmetry. Under KDB supervision, a selected Guarantee Advisor will closely monitor the existing guarantee portfolio so that aggregation of guarantor's then-current liabilities under all executed guarantee agreements does not exceed the total guarantee pool of USD 100 million.

As the Accredited Entity, KDB will visit the Programme operation sites to oversee the status of implementation and technology deployment operation of SMEs in industries on the ground and to supervise the overall process related to financial management. The site visit plan is, however, to materialise once the COVID-19 crisis and ensuing travel restrictions become loose. If the pandemic crisis severely continues in Indonesia, KDB will make its full efforts to secure effective alternatives: hiring local experts to execute the onsite reviews. In addition, KDB plans to monitor and manage FI's implementation activities in line with the guidelines of GCF and KDB through regular meetings with FIs (face-to-face or virtual).

Audit and Internal Control on Proceeds. KDB manages GCF proceeds from GCF to executing entities in line with AMA and FAA, and all the proceeds are subject to internal and external audit in line with KDB rules and guidelines. GCF proceeds from KDB to final beneficiaries are managed by executing entities (EEs) (especially by PIU, which is Programme Implementation Unit established for the Programme for each executing entity) in line with their regulations assessed by KDB already, under KDB's supervision and monitoring.

In addition, all disbursements under Component 1 are controlled by KDB and local FI as following clause of The Guarantee Agreement between KDB and local FIs so that they can be used transparently for the

Programme; each disbursement to the FIs will be based on delivery by the FIs of evidence satisfactory to KDB of the physical progress of Sub-projects. All expenditures or disbursements under Components 2 and 3 are also controlled by KDB and executing entities as follows; each disbursement to the executing entities will be based on delivery by the executing entities of evidence satisfactory to KDB of actual service progress of Sub-projects. Especially, all the disbursement requests should have the details of usage of previous disbursement and their actual documentary evidence such as receipts, invoices, delivery notices, etc. Documentation for usage of disbursement proceeds should be reported on a quarterly basis. Additionally, financial statements of the respective participating entities should be delivered to the entity that it reports to (e.g., IEs to EEs, EEs to AE, and AE to GCF) on an at least semi-annual basis, and audited financial statements and annual reports should be delivered annually for purposes of governance and review.

Procurement Activities in Compliance with Proven KDB Procurement Policies Aligned with the GCF Standard. KDB has a procurement policy that abides by the GCF procurement guidance. All the procurement of goods and services will be undertaken in accordance with KDB's internal and external provision (under supervision of the Korean government) on procurement cleared by the GCF during accreditation. Under KDB supervision, executing entities are all to cling to the guidance.

Under Components 2 and 3, ACE will function as the Programme proponent and pre-selected Executing Entity partner for arranging TA activities. In that, if ACE ought to depend on the expertise of third-party consultants for various activities beyond its own capacity, professional entities will be appointed and/or procured as implementing entities by ACE under KDB's supervision and control.

The procurement process in its entirety will be carried out in strict compliance with KDB's procurement standard as defined in Annex 10 (Procurement Plan) and its respective internal procurement policy.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☐ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☐ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)³⁰
- ☒ Annex 23 Capacity Assessment of Executing Entities
- ☒ Annex 24 Exclusion List

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.